ON THE GEODETIC LINE.

INAUGURAL DISSERTATION

WHICH,

WITH THE CONSENT AND AUTHORITY

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FOR

DULY OBTAINING THE HIGHEST HONORS IN PHILOSOPHY

ON THE FIRST DAY OF MARCH, 1862,
AT THE CUSTOMARY PLACE AND HOUR,

WILL PUBLICLY DEFEND

THE AUTHOR

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BERLIN
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TO HIS MOST BELOVED FATHER

THESE
FIRST FRUITS OF STUDY
WITH A DUTIFUL AND GRATEFUL HEART
HE GIVES, DEDICATES, AND DEVOTES.
THE AUTHOR.

CHAPTER I.

§ 1.

A spheroid is a surface formed by the rotation of an ellipse about its minor axis „b“. In this motion the individual points of the periphery of the ellipse describe circles (parallels), the greatest of which, described by the extreme points of the major axis „a“, is called the equator. —

§ 2.

The minor axis of the ellipse, about which the rotation has been made, is called the axis of the spheroid; its extreme points are the poles; and every plane containing that axis is a meridian.

§ 3.

The angle „ψ“, namely the one formed by the meridian containing the point „A“ and any chosen first meridian, is called the longitude; but the angle „B“, which the normal erected at A on the spheroid makes with the plane of the equator, is the latitude of the point A. Hence one sees that every point of the spheroid is determined by longitude and latitude. —

§ 4.

Since all points of the same longitude ψ lie on the same meridian, and all points of the same latitude B lie on the same circle, we call the angle ψ the longitude of the meridian containing those same points, and the angle B the latitude of the circle containing those same points. —

§ 5.

Besides the latitude B, let us use another angle, „l“, to determine the position of a circle. For, if in place of the radius of the spheroid we use the axis „a“ and describe a sphere about the center, the plane of a circle of the spheroid cuts this sphere also in a circle. Each of the normal lines erected on the sphere at the points of the periphery

of this circle makes with the equator an angle equal to all the others; this angle is called the „reduced latitude“ of the parallel circle of the spheroid and is denoted by the sign „l“. —

§ 6.

Thus every circle of the spheroid, since it is determined both by its latitude „B“ and by the latitude „l“, is called the circle „B“, or the circle „l“. The radius of the circle „l“ is $a \cos l$.

§ 7.

Either latitude is easily determined from the other. In fact, between the angle l and the angle B there is the following relation:

$$a \operatorname{tg} l = b \operatorname{tg} B.$$

Or, since the eccentricity of the meridian ellipse, namely $\sqrt{1 - \frac{b^2}{a^2}}$, is denoted by „$e$“, the relation is:

$$\operatorname{tg} l = \sqrt{1 - e^2} \operatorname{tg} B.$$

Although this is the simplest way of reducing the latitudes B and l one to the other, we shall nevertheless often use other relations, all of which are known to be reducible to the preceding one.

The most important of these relations are:

$$\sin l = \frac{b}{a} \frac{\sin B}{\sqrt{1 - e^2 \sin^2 B}}, \qquad \sin B = \frac{\sin l}{\sqrt{1 - e^2 \cos^2 l}},$$

$$\cos l = \frac{\cos B}{\sqrt{1 - e^2 \sin^2 B}}, \qquad \cos B = \frac{b}{a} \frac{\cos l}{\sqrt{1 - e^2 \cos^2 l}},$$

$$\frac{b^2}{a^2} = 1 - e^2 = (1 - e^2 \sin^2 B)(1 - e^2 \cos^2 l).$$

Finally, putting

$$\frac{a - b}{a + b} = \frac{1 - \sqrt{1 - e^2}}{1 + \sqrt{1 - e^2}} = \varepsilon,$$

we have

$$B = l + \varepsilon \sin 2l + \frac{\varepsilon^2}{2} \sin 4l + \frac{\varepsilon^3}{3} \sin 6l.$$

§ 8.

Since, as we said above, every point of the spheroid is determined by the angles l and ψ, these may be taken as coordinates. Just as the position of any point in a plane is defined by two coordinates, and a plane curve is determined by a relation between those two coordinates, so a relation between l and ψ expresses a curve on the spheroid. Further, as in the discussion of plane curves one usually treats most especially the length and the inclination at each point to some fixed axis

so on the spheroid we wish to consider more precisely the arc „s“ of a curve (which, because ψ is the longitude, we choose to call the extension), and the angle „T“ (azimuth), which the curve makes with the meridian at the points where it cuts the meridian.

§ 9.

It remains, then, to refer the quantities s and T to the quantities l and ψ. Let us therefore consider on some spheroidal curve two points $A(l, \psi)$ and $B(l + dl, \psi + d\psi)$, so close together that no nearer points can be conceived. Let the meridian of the point A cut the circle of the point B in the point $C(l + dl, \psi)$. Since the triangle ABC formed by these points is infinitesimal, it may be taken as lying in a plane. Its sides AC and BC, the one tangent to a meridian and the other to a circle, are inclined at a right angle.

Accordingly,

$$\frac{BC}{AC} = \operatorname{tg} BAC; \qquad \frac{AC}{AB} = \cos BAC; \qquad \frac{BC}{AB} = \sin BAC,$$

and

$$AB^2 = AC^2 + BC^2.$$

But

$$BAC = T; \qquad AB = ds,$$

$$BC \text{ (the arc of a circle described with radius „}a \cos l\text{“)} = a \cos l \, d\psi,$$

$$AC \text{ (the arc of the ellipse)} = a\sqrt{1 - e^2 \cos^2 l}\, dl.$$

Putting these values in place of the preceding ones, we obtain

$$ds^2 = a^2(1 - e^2 \cos^2 l)\, dl^2 + a^2 \cos^2 l \, d\psi^2,$$

$$\operatorname{tg} T = \frac{\cos l}{\sqrt{1 - e^2 \cos^2 l}} \frac{d\psi}{dl}; \qquad \cos T = dl \sqrt{\frac{1 - e^2 \cos^2 l}{(1 - e^2 \cos^2 l) dl^2 + \cos^2 l \, d\psi^2}},$$

$$\sin T = d\psi \sqrt{\frac{\cos^2 l}{(1 - e^2 \cos^2 l) dl^2 + \cos^2 l \, d\psi^2}}.$$

§ 10.

For the earth, from the calculations proposed by the illustrious Bessel, the following ratios are relevant:

$$\begin{aligned}
a &= 3272077,1399 \text{ toises}; & b &= 3261139,3284 \text{ toises},\\
\log a &= 6,5148235337; & \log b &= 6,5133693539,\\
\log \tfrac{b}{a} &= 9,9985458202; & \log \tfrac{a}{b} &= 0,0014541798,\\
\varepsilon = \tfrac{a-b}{a+b} &= \tfrac{1-\sqrt{1-e^2}}{1+\sqrt{1-e^2}} = 0,001674184767, & &\\
\log e &= 8,9122052075; & \log \varepsilon &= 7,2238033861.
\end{aligned}$$

CHAPTER II.

§ 1.

A curve described by the point Π on the spheroid, while no force acts on it except a normal force, is called a geodetic line. We find its equation by eliminating from the equations of motion of the moving point the quantities t and λ, of which the latter represents the normal force and the former a proportional time.

§ 2.

Taking, therefore, the equation of the spheroid as „$f = 0$“, the equations of motion in rectangular coordinates are

$$\frac{d_2x}{dt^2} = \lambda\frac{\partial f}{\partial x}, \qquad \frac{d_2y}{dt^2} = \lambda\frac{\partial f}{\partial y}, \qquad \frac{d_2z}{dt^2} = \lambda\frac{\partial f}{\partial z}.$$

If these are successively multiplied by the differentials

$$dx \qquad dy \qquad dz,$$

and added, there results

$$\frac{dx}{dt}\frac{d_2x}{dt^2} + \frac{dy}{dt}\frac{d_2y}{dt^2} + \frac{dz}{dt}\frac{d_2z}{dt^2} = 0;$$

or, on integrating,

$$\frac{dx}{dt}\frac{dx}{dt} + \frac{dy}{dt}\frac{dy}{dt} + \frac{dz}{dt}\frac{dz}{dt} = \frac{1}{C^2};$$

that is,

$$dt = C\,ds.$$

Thus we see the familiar fact that the velocity of the point Π is constant.

§ 3.

Introducing this value into the equations of motion, we have

$$\frac{d_2x}{ds^2} = \lambda C^2\frac{\partial f}{\partial x}, \qquad \frac{d_2y}{ds^2} = \lambda C^2\frac{\partial f}{\partial y}, \qquad \frac{d_2z}{ds^2} = \lambda C^2\frac{\partial f}{\partial z}.$$

These express nothing else than this: the variation of the extension, while the relation $f = 0$ is maintained in forming it, itself vanishes.

Therefore the equation of the geodetic line is

$$\delta \int \left\{ \sqrt{dx^2 + dy^2 + dz^2} + \lambda f \right\} = 0,$$

or, if instead of the rectangular coordinates „x, y, z“ we use the elliptic coordinates l, ψ,

$$\delta \int \sqrt{(1 - e^2 \cos l)\, dl^2 + \cos^2 l\, d\psi^2} = 0.$$

§ 4.

Here, if we apply the customary methods of analysts, one finally obtains

$$\frac{\cos^2 l \, \frac{d\psi}{dl}}{\sqrt{1 - e^2 \cos^2 l + \cos^2 l \, \frac{d\psi}{dl} \frac{d\psi}{dl}}} = \text{Const.}$$

Using in this formula

$$\frac{\cos l \, \frac{d\psi}{dl}}{\sqrt{1 - e^2 \cos^2 l + \cos^2 l \, \frac{d\psi}{dl} \frac{d\psi}{dl}}} = \sin T$$

compare Chapter I, § 9,

we have

$$\cos l \sin T = \text{Const.}$$

And if, taking

$$T = \frac{\pi}{2}, \qquad l = l_0,$$

we put

$$\cos l \sin T = \cos l_0,$$

then

$$\cos l \cos T = \sqrt{\sin^2 l_0 - \sin^2 l}, \qquad \operatorname{tg} T = \frac{\cos l_0}{\sqrt{\sin^2 l_0 - \sin^2 l}}.$$

§ 5.

That is: a geodetic line which touches some circle also touches another circle having the corresponding position. Further, since always

$$l_0 > l,$$

it can never happen that the geodetic line recedes from the equator beyond these circles. Therefore we call the geodetic line tangent to the circle l_0 the geodetic line l_0, and the point at which the circle is touched the origin of the geodetic line; from this origin we shall henceforth measure the extensions „s“ and the longitudes ψ of the points.

§ 6.

To determine these from the latitude l, applying formulas II to the formulas proposed in Chapter I, § 9, we have

$$a\cos l\,dl\sqrt{1-e^2\cos^2 l} = -\sqrt{\sin^2 l_0 - \sin^2 l}\,ds,$$
$$\cos l_0\,dl\sqrt{1-e^2\cos^2 l} = \mp\sqrt{\sin^2 l_0 - \sin^2 l}\,\cos l\,d\psi.$$

In the latter equation the negative or positive sign is to be used according as the longitudes, when the point Π recedes from the origin, increase or decrease. On integrating, there results

$$s = -a\int_0^l \frac{dl\cos l\sqrt{1-e^2\cos^2 l}}{\sqrt{\sin^2 l_0 - \sin^2 l}},$$
$$\psi = \mp\int_0^l \frac{dl\cos l_0\sqrt{1-e^2\cos^2 l}}{\cos l\sqrt{\sin^2 l_0 - \sin^2 l}}.$$

§ 7.

If the limit l is defined so that it has a value greater than „$-l_0$“ and less than „$+l_0$“, then from these formulas there are obtained single appropriate values of both extension and longitude. This agrees with the assertion that the point Π in its motion cuts the circle l at only one place; but because the point moves perpetually with constant velocity, this cannot be so. For this reason the formulas proposed by us belong not to the whole extension of the geodetic line l_0, but to that part which lies between the origin and the point at which the circle $-l_0$ is touched.

Therefore, in order that they may be transformed into formulas of the general kind, in place of the latitude l, which up to now we have taken as variable by itself, there must be introduced a variable of such a kind that it increases according to the recession of the point Π from the origin. Such a variable is the angle φ, defined by the equation

$$\sin l_0\cos\varphi = \sin l.$$

§ 8.

The angle φ rightly contains the qualities that we are seeking. For the relation between the geodetic line and the angle φ is this: at the origin $\varphi = 0$; at the section with the equator $\varphi = \frac{\pi}{2}$; at the tangency with the circle $-l_0$, $\varphi = \pi$; at the second section with the equator $\varphi = \frac{3}{2}\pi$; at the second tangency with the circle l_0, $\varphi = 2\pi$, and so on. Similarly, as the line proceeds, when it touches either circle at those points which we choose to call points of conversion, $\varphi = m\pi$; when it cuts the equator, $\varphi = \frac{2m+1}{2}\pi$.

§ 9.

Since to each point of the geodetic line l_0 there correspond certain values of the angle φ, the quantities

$$l \qquad B \qquad T \qquad s \qquad \psi$$

must necessarily be expressed as functions of the angle φ. The quantity l is determined from the equation

$$\sin l = \sin l_0 \cos\varphi,$$

or from these:

$$\cos l = \sqrt{1-\sin^2 l_0\cos^2\varphi} = \cos l_0\sqrt{1+\operatorname{tg}^2 l_0\sin^2\varphi},$$
$$\operatorname{tg} l = \frac{\operatorname{tg} l_0\cos\varphi}{\sqrt{1+\operatorname{tg}^2 l_0\sin^2\varphi}} = \frac{\cos\varphi}{\sqrt{\cot^2 l_0+\sin^2\varphi}},$$
$$\sqrt{\sin^2 l_0-\sin^2 l} = \sin l_0\sin\varphi,$$
$$\sqrt{1-e^2\cos^2 l} = \sqrt{1-e^2+e^2\sin^2 l_0\cos^2\varphi} = \sqrt{1-e^2\cos^2 l_0-e^2\sin^2 l_0\sin^2\varphi}.$$

Substituting these values into the equations proposed in Chapter I, § 7, and Chapter II, § 4 and § 6, we obtain

$$\sqrt{1-e^2}\ \operatorname{tg} B = \frac{\cos\varphi}{\sqrt{\cot^2 l_0+\sin^2\varphi}},$$

$$\sin B = \frac{\sin l_0\cos\varphi}{\sqrt{1-e^2+e^2\sin^2 l_0\cos^2\varphi}},$$

$$\cos B = \cos l_0\sqrt{1-e^2}\sqrt{\frac{1+\operatorname{tg}^2 l_0\sin^2\varphi}{1-e^2+e^2\sin^2 l_0\cos^2\varphi}},$$

$$\operatorname{tg} T = \frac{\cot l_0}{\sin\varphi}, \qquad \sin T = \frac{1}{\sqrt{1+\operatorname{tg}^2 l_0\sin^2\varphi}}, \qquad \cos T = \frac{\sin\varphi}{\sqrt{\cot^2 l_0+\sin^2\varphi}},$$

$$s = a\int_0^\varphi d\varphi\sqrt{1-e^2+e^2\sin^2 l_0\cos^2\varphi} = a\int_0^\varphi d\varphi\sqrt{1-e^2\cos^2 l_0-e^2\sin^2 l_0\sin^2\varphi},$$

$$\psi = \pm\int_0^\varphi d\varphi\,\frac{\cos l_0}{\sin^2 l_0}\,\frac{\sqrt{1-e^2\cos^2 l_0-e^2\sin^2 l_0\sin^2\varphi}}{\cot^2 l_0+\sin^2\varphi}.$$

The last two equations, putting

$$\frac{e\sin l_0}{\sqrt{1-e^2\cos^2 l_0}} = e\sin B_0 = x,$$

and

$$\sqrt{1-e^2\sin^2 B_0} = \frac{\cos B_0}{\cos l_0} = x',$$

are transformed into

$$s = \frac{b}{x'}\int_0^\varphi d\varphi\sqrt{1-x^2\sin^2\varphi},$$

$$\psi = \pm\frac{\cot l_0}{\sin B_0}\int_0^\varphi d\varphi\,\frac{\sqrt{1-x^2\sin^2\varphi}}{\cot^2 l_0+\sin^2\varphi}.$$

CHAPTER III.

§ 1.

Let there be named:

$$a\int_0^{\frac{\pi}{2}} d\varphi\sqrt{1-e^2+e^2\sin^2 l_0\cos^2\varphi}$$

the period of extension, and

$$\int_0^{\frac{\pi}{2}} \cos l_0\, d\varphi\, \frac{\sqrt{1-e^2+e^2\sin^2 l_0\cos^2\varphi}}{1-\sin^2 l_0\cos^2\varphi}$$

the period of longitude of the geodetic line l_0; since these depend only on the angle l_0, let them be denoted by $\Sigma(l_0)$ and $\mathcal{A}(l_0)$. Since the values of these periods will hereafter be of very great importance, the series must now be developed by which they are computed easily and with the greatest possible accuracy.

§ 2.

For finding $\Sigma(l_0)$ we expand under the sign of integration

$$\sqrt{1-e^2+e^2\sin^2 l_0\cos^2\varphi}$$

according to the binomial theorem, where, for brevity, we put:

$$\frac{1}{4}\,\frac{e^2}{1-e^2} = \frac{\varepsilon}{(1-\varepsilon)^2} = o.$$

It then follows:

$$\begin{aligned}\frac{1}{b}\Sigma(l_0) &= \int_0^{\frac{\pi}{2}} d\varphi\{1+2o\sin^2 l_0\cos^2\varphi - 2o^2\sin^4 l_0\cos^4\varphi \\ &\qquad + 4o^3\sin^6 l_0\cos^6\varphi - 10o^4\sin^8 l_0\cos^8\varphi\ldots\} \\ &= \int_0^{\frac{\pi}{2}} d\varphi + 2o\sin^2 l_0\int_0^{\frac{\pi}{2}}\cos^2\varphi\, d\varphi \\ &\quad - 2o^2\sin^4 l_0\int_0^{\frac{\pi}{2}}\cos^4\varphi\, d\varphi + 4o^3\sin^6 l_0\int_0^{\frac{\pi}{2}}\cos^6\varphi\, d\varphi \\ &\quad - 10o^4\sin^8 l_0\int_0^{\frac{\pi}{2}}\cos^8\varphi\, d\varphi\ldots\end{aligned}$$

The individual integrals have this form:

$$\int_0^{\frac{\pi}{2}}\cos^{2m}\varphi\, d\varphi.$$

They therefore are equal to:

$$\frac{\pi}{2^{2m+1}} \cdot \frac{2m \cdot (2m-1) \cdot (2m-2) \cdots (m+1)}{1 \cdot 2 \cdot 3 \cdots m}.$$

If these values are reduced to our formula, there results:

$$\frac{2}{b\pi}\Sigma(l_0) = 1 + o\sin^2 l_0 - \frac{3}{4}o^2\sin^4 l_0 + \frac{5}{4}o^3\sin^6 l_0 - \frac{175}{64}o^4\sin^8 l_0 \ldots.$$

§ 3.

From this you see that, when

$$\Sigma(l_0') > \Sigma(l_0''),$$

one has

$$l_0' > l_0'',$$

which agrees with the fact that l_0 is a singly fitted function of the period of extension, or that a geodetic line is completely determined by its period of extension.

Therefore, by the inverse series and putting

$$\frac{2}{b\pi} \cdot \Sigma(l_0) - 1 = u,$$

there exists:

$$o\sin^2 l_0 = u + \frac{3}{4}u^2 - \frac{1}{8}u^3 + \frac{5}{32}u^4.$$

§ 4.

To unravel the period of longitude, computing

$$\frac{d\Sigma(l_0)}{dl_0}$$

and using the identity

$$\frac{d\Sigma(l_0)}{dl_0} = a\cos l_0 \, \frac{d\mathcal{A}(l_0)}{dl_0},$$

we obtain

$$\begin{aligned} a\cos l_0 \, \frac{d\mathcal{A}(l_0)}{dl_0} &= \frac{b}{2} \cdot \pi\cos l_0\{2o\sin l_0 - 3o^2\sin^3 l_0 + \frac{15}{2}o^3\sin^5 l_0 \\ &\qquad - \frac{175}{8}o^4\sin^7 l_0 \ldots\} \\ \frac{d\mathcal{A}(l_0)}{dl_0} &= \frac{b}{a} \cdot \frac{\pi}{2}\sin l_0\{2o - 3o^2\sin^2 l_0 + \frac{15}{2}o^3\sin^4 l_0 \\ &\qquad - \frac{175}{8}o^4\sin^6 l_0 \ldots\} \\ &= \frac{b\pi}{2a}\sin l_0\{(2 - 3o + \frac{15}{2}o^2 - \frac{175}{8}o^3)o + (3 - 15o + \frac{525}{8}o^2)o^2\cos^2 l_0 \\ &\qquad + (\frac{15}{2} - \frac{525}{8}o)o^3\cos^4 l_0 + \frac{175}{8}o^4\cos^6 l_0 \ldots\}. \end{aligned}$$

And integrating between the limits 0 and $\frac{\pi}{2}$,

$$\begin{aligned} \frac{\pi}{2} - \mathcal{A}(l_0) = \frac{b}{a}\frac{\pi}{2}\{&(2 - 3o + \frac{15}{2}o^2 - \frac{175}{8}o^3)o\cos l_0 + (1 - 5o + \frac{175}{8}o^2)o^2\cos^3 l_0 \\ &+ (\frac{3}{2} - \frac{105}{8}o)o^3\cos^5 l_0 + \frac{25}{8}o^4\cos^7 l_0, \ldots\}. \end{aligned}$$

In this series the constant coefficients of the individual powers of the function „$\cos l_0$“ are brought to simpler forms by putting:

$$p = o - \frac{3}{2}o^2 + \frac{15}{4}o^3 - \frac{175}{8}o^4,$$

or, by the inverse series,

$$o = p + \frac{3}{2}p^2 + \frac{3}{4}p^3 - \frac{5}{16}p^4.$$

When this series is substituted for o in the equation, there results:

$$\frac{2\mathcal{A}(l_0)}{\pi} = 1 - \frac{b}{a}\{2p\cos l_0 + (1 - 2p + \frac{25}{8}p^2)p^2\cos^3 l_0 + (\frac{3}{2} - \frac{51}{8}p)p^3\cos^5 l_0 + \frac{25}{8}p^4\cos^7 l_0 \ldots\}.$$

§ 5.

Just as the definition of the geodetic line is sought from the period of extension, so it is sought from the period of longitude.

Accordingly, from $\mathcal{A}(l_0)$ the value l_0 can be computed explicitly; putting

$$a\left\{1 - \frac{2\mathcal{A}(l_0)}{\pi}\right\} = 2bv\sqrt{p} \qquad \text{and} \qquad \cos l_0\sqrt{p} = x,$$

one has:

$$v = x + x^3\left(\frac{1}{2} - p + \frac{25}{16}p^2\right) + x^5\left(\frac{3}{4} - \frac{51}{16}p\right) + \frac{25}{16}x^7 \ldots$$

and, if we invert this equation,

$$x = v + v^3\left(-\frac{1}{2} + p - \frac{25}{16}p^2\right) + v^5\left(\frac{3}{16}p - \frac{33}{32}p^2\right) - \frac{1}{16}v^7.$$

§ 6.

Using the numbers explained in the first chapter, the forms are these:

$$o = 0{,}001\ 6798\ 046\ 65 \qquad p = 0{,}001\ 1755\ 897\ 26$$

$$\Sigma(l_0) = \frac{b\pi}{2} + \frac{b\pi}{2\cdot 10^{12}}\{167\ 9804\ 665\sin^2 l_0 - 211\ 6291\sin^4 l_0 + 5920\sin^6 l_0 - 20\sin^8 l_0 \ldots\}$$

$$\mathcal{A}(l_0) = \frac{\pi}{2} - \frac{b\pi}{2\cdot 10^{12}}\{335\ 1179\ 452\cos l_0 + 282\ 6309\cos^3 l_0 + 7125\cos^5 l_0 + 22\cos^7 l_0 \ldots\}.$$

CHAPTER IV.

§ 1.

The integral

$$\int_{l_1}^{l_2} dl\sqrt{1 - e^2\cos^2 l + \cos^2 l\,\frac{d\psi}{dl}\cdot\frac{d\psi}{dl}}$$

which, as we taught above, signifies the extension between the points $A(l_1, \psi_1)$ and $B(l_2, \psi_2)$ of any line whatsoever on the spheroid, becomes a minimum in a certain line.

The relations pertaining to it are these:

$$\delta_1 \int_{l_1}^{l_2} dl \sqrt{1 - e^2 \cos^2 l + \cos^2 l \, \frac{d\psi}{dl} \cdot \frac{d\psi}{dl}} = 0$$

$$\delta_2 \int_{l_1}^{l_2} dl \sqrt{1 - e^2 \cos^2 l + \cos^2 l \, \frac{d\psi}{dl} \cdot \frac{d\psi}{dl}}$$
$$= \int_{l_1}^{l_2} \left(\frac{dl \, \delta\left(\frac{d\psi}{dl}\right) \delta\left(\frac{d\psi}{dl}\right) \cos^2 l [1 - e^2 \cos^2 l]}{\{1 - e^2 \cos^2 l + \cos^2 l \, \frac{d\psi}{dl} \cdot \frac{d\psi}{dl}\}^{1\frac{1}{2}}} \right) > 0.$$

This line, since it is shorter than neighboring lines, shall be called shortest.

And, lest perhaps an error be introduced by this name, let us call that line, than which no shorter line at all can be conceived, the shortest of all; this is to be understood as a part of the shortest line, while this part itself is a part of the geodetic.

§ 2.

Since the integral

$$\int_{l_1}^{l_2} \left(\frac{dl \, \delta\left(\frac{d\psi}{dl}\right) \delta\left(\frac{d\psi}{dl}\right) \cos^2 l [1 - e^2 \cos^2 l]}{\{1 - e^2 \cos^2 l + \cos^2 l \, \frac{d\psi}{dl} \cdot \frac{d\psi}{dl}\}^{1\frac{1}{2}}} \right)$$

and each element of it are known to be positive, the whole integral must be positive. It would seem to agree with this that every geodetic line, even in its entire extension, is shortest.

This is not so, because in Chapter II, § 7, we showed that the force of our equation holds only up to the point at which the circle „$-l_0$“ is touched by the geodetic line.

§ 3.

Therefore, since from the integral equation which we proposed above one cannot know how far a geodetic line, even after touching the circle „$-l_0$“, is shortest, universal rules must be established. We shall obtain this by introducing, in place of the variable of integration l, another variable of integration, ψ, which evidently increases perpetually in proportion as the point Π recedes from the origin.

Accordingly, it must be asked what is the greatest longitude ψ up to which, perpetually,

$$\delta_2 \int_{\psi_1}^{\psi_2} d\psi \sqrt{(1 - e^2 \cos^2 l) l'^2 + \cos^2 l}$$

$$\begin{aligned}
&\delta l'\delta l' \cos^2 l(1-e^2\cos^2 l) + 2\delta l\delta l' \sin l\cos l\, l'\big(e^2 l'^2[1-e^2\cos^2 l] + [1+e^2\cos^2 l]\big)\\
=&\int_{\psi_1}^{\psi_2} d\psi \,\frac{+\delta l^2\big(e^2 l'^4[\cos 2l - e^2\cos^4 l] + l'^2[2e^2\cos^4 l - 2\cos 2l]\big)}{\{(1-e^2\cos^2 l)l'^2 + \cos^2 l\}^{1\frac{1}{2}}}\\
=&\int_{\psi_1}^{\psi_2} d\psi\{(11)\delta l'\delta l' + 2(01)\delta l\delta l' + (00)\delta l\delta l\} > 0
\end{aligned}$$

where, for this integral, $\delta l = 0$ at the limits.

§ 4.

The elements of this integral, for values assigned to the variation δl, can become either positive or negative. Whether the whole integral is positive or negative is determined by this, whether the sum of the positive elements or of the negative elements is the greater. Since, however, the question is how far our integral is positive for values of integration of whatever kind, the whole question must be referred to this: how far an integral is found which, since it is itself 0, when added to that integral, makes all its elements positive in force.

§ 5.

An integral of this kind is the following:

$$\int_{\psi_1}^{\psi_2} d\psi\{\delta l\delta l N\}' = \int_{\psi_1}^{\psi_2} d\psi\{2N\delta l\delta l' + N'\delta l\delta l\}.$$

This is so in so far as the function N can be defined in such a way that the sum

$$(11)\delta l'\delta l' + 2[(10)+N]\delta l\delta l' + [(00)+N']\delta l\delta l$$

throughout the whole interval, with N never becoming infinite, becomes a complete square.

For under this condition

$$\int_{\psi_1}^{\psi_2} d\psi\{2N\delta l\delta l' + N'\delta l\delta l\} = N_2\delta l_2\delta l_2 - N_1\delta l_1\delta l_1 = 0$$

since the variations vanish at the limits.

§ 6.

The sum

$$(11)\delta l'\delta l' + 2[(10)+N]\delta l\delta l' + [(00)+N']\delta l\delta l$$

then is a square when

$$(11)[(00)+N'] = [(01)+N]^2,$$

or, integrating,

$$-N = (01) + (11)\frac{\alpha\frac{\partial l'}{\partial\psi_1} + \beta\frac{\partial l'}{\partial l_0}}{\alpha\frac{\partial l}{\partial\psi} + \beta\frac{\partial l}{\partial l_0}}.$$

In this equation α and β are arbitrary constants.

The integral

$$\int_{\psi_1}^{\psi_2} d\psi\{(11)\delta l'\delta l' + 2(01)\delta l\delta l' + (00)\delta l\delta l\}$$

is therefore positive so long as the constants α and β can be determined in such a way that N never becomes infinite in the interval from ψ_1 to ψ_2.

§ 7.

Accordingly it is useful to transform the function N.

For this purpose we differentiate the equation of the geodetic line:

$$\psi - \psi_1 = \int_{\varphi_1}^{\varphi} \frac{d\varphi \cos l_0 \sqrt{1 - e^2 + e^2 \sin^2 l_0 \cos^2 \varphi}}{1 - \sin^2 l_0 \cos^2 \varphi}$$

(in which the angles φ_1 and φ are known to be defined thus:

$$\sin l_0 \cos \varphi_1 = \sin l, \qquad \sin l_0 \cos \varphi = \sin l)$$

partially with respect to ψ_1 and l_0, with ψ to be taken as constant in the differentiation.

Hence the identities follow:

$$-1 = \frac{\cos l_0 \sqrt{1 - e^2 + e^2 \sin^2 l_0 \cos^2 \varphi}}{1 - \sin^2 l_0 \cos^2 \varphi} \frac{d\varphi}{d\psi_1} = -\cot l_0 \frac{\sqrt{1 - e^2 \cos^2 l}}{\cos l \sin \varphi} \frac{dl}{d\psi_1}$$

$$0 = e^2 \sin^2 l_0 \int_{\varphi_1}^{\varphi} d\varphi \frac{\cos^2 \varphi}{\sqrt{1 - e^2 + e^2 \sin^2 l_0 \cos^2 \varphi}} + \cot \varphi \sqrt{1 - e^2 \cos^2 l} - \cot \varphi_1 \sqrt{1 - e^2 \cos^2 l_1}$$
$$- \cos l_0 \frac{\sqrt{1 - e^2 \cos^2 l}}{\cos l \sin \varphi} \cdot \frac{dl}{dl_0},$$

from which, putting

$$e^2 \sin^2 l_0 \sin \varphi \int_{\varphi_1}^{\varphi} d\varphi \frac{\cos^2 \varphi}{\sqrt{1 - e^2 + e^2 \sin^2 l_0 \cos^2 \varphi}}$$
$$+ \cos \varphi \sqrt{1 - e^2 \cos^2 l} - \cot \varphi_1 \sin \varphi \sqrt{1 - e^2 \cos^2 l_1} = u.$$

$$\frac{\cos l_0}{\cos l} \sqrt{1 - e^2 \cos^2 l} = v,$$

the quotients of the differentials are found:

$$\frac{dl}{d\psi} = \frac{\sin l_0 \sin \varphi}{v}; \qquad \frac{dl}{dl_0} = \frac{u}{v}.$$

When these values are carried back to N, there results:

$$-N = (01) + (11) \frac{\left(\alpha \frac{\sin l_0 \sin \varphi}{v} + \beta \frac{u}{v}\right)'}{\alpha \frac{\sin l_0 \sin \varphi}{v} + \beta \frac{u}{v}}$$
$$= (01) + \frac{(11) \cdot (1 - \sin^2 l_0 \cos^2 \varphi)}{\cos l_0 \sqrt{1 - e^2 + e^2 \sin^2 l_0 \cos^2 \varphi}} \cdot \frac{v \frac{d}{d\varphi} \left\{\alpha \frac{\sin l_0 \sin \varphi}{v} + \beta \frac{u}{v}\right\}}{\alpha \sin l_0 \sin \varphi + \beta u}.$$

$$= (10) + \frac{(11)\cos^2 l}{\cos l_0 \sqrt{1 - e^2 \cos^2 l}} \left\{ \frac{\alpha \sin l_0 \cos\varphi + \beta \frac{du}{d\varphi}}{\alpha \sin l_0 \sin\varphi + \beta u} - \frac{\frac{dv}{d\varphi}}{v} \right\}.$$

§ 8.

Since on the right side of the equation the quantities

$$(01) \quad (11) \quad \cos^2 l \quad \operatorname{tg} l_0 \quad \frac{1}{\sqrt{1 - e^2 \cos^2 l}} \quad \cos\varphi \quad \frac{du}{d\varphi} \quad \frac{dv}{d\varphi} \quad \frac{1}{v}$$

remain perpetually finite, the function N can become infinite only in this way, namely if

$$\alpha \sin l_0 \sin\varphi + \beta u = 0$$

holds.

It follows from this that the integral

$$\int_{\psi_1}^{\psi_2} d\varphi \{(11)\delta l' \delta l' + 2(01)\delta l \delta l' + (00)\delta l \delta l\}$$

is positive so long as the constants α and β can be determined in such a way that

$$\alpha \sin l_0 \sin\varphi + \beta u$$

does not vanish; thus so long as

$$\frac{u}{\sin l_0 \sin\varphi}$$

does not assume all numbers lying between $-\infty$ and $+\infty$. For if

$$\frac{u}{\sin l_0 \sin\varphi}$$

does not assume some value "a" in this interval, then, putting

$$\alpha + \beta a = 0,$$

there is never

$$\alpha \sin l_0 \sin\varphi + \beta u = 0.$$

§ 9.

Since

$$\frac{d}{d\varphi} \left\{ \frac{u}{\sin l_0 \sin\varphi} \right\} = -\frac{1 - \sin^2 l_0 \cos^2\varphi}{\sin l_0 \cos l_0 \sin^2\varphi}$$

is always negative, and since, on putting $\psi = \psi_1$,

$$\frac{u}{\sin l_0 \sin\varphi} = 0,$$

as ψ increases,

$$\frac{u}{\sin l_0 \sin\varphi}$$

assumes in succession all values from 0 down to $-\infty$ and, after passing back from $-\infty$ to $+\infty$, from ∞ to 0.

So long as this path has not yet been traversed, α and β can be determined so that the function N remains finite; but when

$$\frac{u}{\sin l_0 \sin \varphi}$$

has gone beyond 0, it is necessary that somewhere in the interval N become infinite, where that rule under which

$$\int_{\psi_1}^{\psi} d\psi\{(11)\delta l'\delta l' + 2(01)\delta l\delta l' + (00)\delta l\delta l\} > 0$$

ceases to hold.

If we recall that

$$\frac{u}{\sin l_0 \sin \varphi}$$

arose by differentiating the equation of the geodetic line partially with respect to ψ according to the constant l_0, in such a way that l is understood as constant, then the following theorem corresponds to what we proposed in the preceding paragraph.

> "A geodetic line is a shortest line so long as it does not cut another neighboring geodetic line issuing from the same point."

The point of this intersection is called the terminus of the geodetic line and is denoted by

$$O(l, \psi).$$

§ 11.

Indeed,

$$\int_{\psi_1}^{\psi_2} d\psi\{(11)\delta l'\delta l' + 2(01)\delta l\delta l' + (00)\delta l\delta l\}$$

(with $\psi = \psi_2$ put) can have δl determined in such a way that it vanishes.

Namely, by putting

$$\delta l = \alpha \frac{\partial l}{\partial l_0},$$

our integral takes this form:

$$\alpha^2 \int_{\psi_1}^{\psi_2} d\psi \left\{(11)\frac{\partial l'}{\partial l_0} \cdot \frac{\partial l'}{\partial l_0} + 2(01)\frac{\partial l}{\partial l_0} \cdot \frac{\partial l'}{\partial l_0} + (00)\frac{\partial l}{\partial l_0} \cdot \frac{\partial l}{\partial l_0}\right\}$$

or the identity

$$\begin{aligned} 0 = &\left((10)\frac{\partial l}{\partial l_0}\frac{\partial l}{\partial l_0} + (11)'\frac{\partial l}{\partial l_0}\frac{\partial l'}{\partial l_0}\right) - \left((01)\frac{\partial l}{\partial l_0}\frac{\partial l'}{\partial l_0} + (11)\frac{\partial l}{\partial l_0}\frac{\partial l'}{\partial l_0}\right), \\ = &\alpha^2 \int_{\psi_1}^{\psi_2} d\psi \left\{-(01)'\frac{\partial l}{\partial l_0}\frac{\partial l}{\partial l_0} - (11)'\frac{\partial l}{\partial l_0}\frac{\partial l'}{\partial l_0} - 2(01)\frac{\partial l}{\partial l_0}\frac{\partial l'}{\partial l_0} - (11)\frac{\partial l'}{\partial l_0}\frac{\partial l'}{\partial l_0} - (11)\frac{\partial l}{\partial l_0}\frac{\partial l''}{\partial l_0}\right\} \end{aligned}$$

adding, this:

$$= \alpha^2 \int_{\psi_1}^{\psi_2} d\psi \left\{ (00)\frac{\partial l}{\partial l_0}\frac{\partial l}{\partial l_0} - (01)'\frac{\partial l}{\partial l_0}\frac{\partial l}{\partial l_0} - (11)'\frac{\partial l}{\partial l_0}\frac{\partial l'}{\partial l_0} - (11)\frac{\partial l}{\partial l_0}\frac{\partial l''}{\partial l_0} \right\}$$

$$= \alpha^2 \int_{\psi_1}^{\psi_2} d\psi \, \frac{\partial l}{\partial l_0} \left\{ \frac{\partial l}{\partial l_0}\big((00) - (01)'\big) - \left((11)\frac{\partial l'}{\partial l_0}\right)' \right\}$$

$$= \alpha^2 \int_{\psi_1}^{\psi_2} d\psi \, \frac{\partial l}{\partial l_0}\frac{\partial}{\partial l_0}\big((0) - (1)'\big) = 0.$$

Hence this is obtained:

> "Geodetic lines having the same extreme points differ in their lengths by quantities that are minima of the third degree."

§ 12.

After we have therefore understood that, with δl suitably defined and with the geodetic line produced as far as the terminus O,

$$U = \int_{\psi_1}^{\psi_2} d\psi \{(11)\delta l'\delta l' + 2(01)\delta l\delta l' + (00)\delta l\delta l\} = 0$$

can be made to hold, it remains to show that, if the line is produced farther,

$$U < 0$$

can exist.

Accordingly we divide our integral into two parts, so that one part is contained in the interval from ψ_1 to ψ_2, the other from ψ_2 to ψ, putting

$$\psi_1 < \psi_2 < \psi.$$

In the first part we take

$$\delta l = \alpha_1 \frac{dl}{d\psi_1} + \beta_1 \frac{dl}{dl_0},$$

and in the latter

$$\delta l = \alpha_2 \frac{dl}{d\psi_1} + \beta_2 \frac{dl}{dl_0},$$

where in these formulae

$$\alpha_1 \qquad \alpha_2 \qquad \beta_1 \qquad \beta_2$$

are arbitrary constants. Hence there results:

$$U = \begin{cases} \displaystyle\int_{\psi_1}^{\psi_2} d\psi \left\{ (11)\left(\alpha_1\frac{dl'}{d\psi_1} + \beta_1\frac{dl'}{dl_0}\right)^2 + 2(01)\left(\alpha_1\frac{dl'}{d\psi_1} + \beta_1\frac{dl'}{dl_0}\right)\left(\alpha_1\frac{dl}{d\psi_1} + \beta_1\frac{dl}{dl_0}\right) + (00)\left(\alpha_1\frac{dl}{d\psi_1} + \beta_1\frac{dl}{dl_0}\right)^2 \right\} \\ \displaystyle\int_{\psi_2}^{\psi} d\psi \left\{ (11)\left(\alpha_2\frac{dl'}{d\psi_1} + \beta_2\frac{dl'}{dl_0}\right)^2 + 2(01)\left(\alpha_2\frac{dl'}{d\psi_1} + \beta_2\frac{dl'}{dl_0}\right)\left(\alpha_2\frac{dl}{d\psi_1} + \beta_2\frac{dl}{dl_0}\right) + (00)\left(\alpha_2\frac{dl}{d\psi_1} + \beta_2\frac{dl}{dl_0}\right)^2 \right\} \end{cases}$$

and, integrating as in § 11, and marking the functions

$$\frac{dl}{d\psi_1} \qquad \frac{dl}{dl_0} \qquad \frac{dl'}{d\psi_1} \qquad \frac{dl'}{dl_0} \qquad (01) \quad (11)$$

by the same subscripts as the corresponding ψ's:

$$U = \begin{cases} (01)_2 \left\{ \alpha_1 \left(\frac{dl}{d\psi_1}\right)_2 + \beta_1 \left(\frac{dl}{dl_0}\right)_2 \right\}^2 + (11)_2 \left\{ \alpha_1 \left(\frac{dl}{d\psi_1}\right)_2 + \beta_1 \left(\frac{dl}{dl_0}\right)_2 \right\} \left\{ \alpha_1 \left(\frac{dl'}{d\psi_1}\right)_2 + \beta_1 \left(\frac{dl'}{dl_0}\right)_2 \right\} \\ - (01)_1 \left\{ \alpha_1 \left(\frac{dl}{d\psi_1}\right)_1 + \beta_1 \left(\frac{dl}{dl_0}\right)_1 \right\}^2 - (11)_1 \left\{ \alpha_1 \left(\frac{dl}{d\psi_1}\right)_1 + \beta_1 \left(\frac{dl}{dl_0}\right)_1 \right\} \left\{ \alpha_1 \left(\frac{dl'}{d\psi_1}\right)_1 + \beta_1 \left(\frac{dl'}{dl_0}\right)_1 \right\} \\ + (01) \left\{ \alpha_2 \left(\frac{dl}{d\psi_1}\right) + \beta_2 \left(\frac{dl}{dl_0}\right) \right\}^2 + (11) \left\{ \alpha_2 \left(\frac{dl}{d\psi_1}\right) + \beta_2 \left(\frac{dl}{dl_0}\right) \right\} \left\{ \alpha_2 \left(\frac{dl'}{d\psi_1}\right) + \beta_2 \left(\frac{dl'}{dl_0}\right) \right\} \\ - (01)_2 \left\{ \alpha_2 \left(\frac{dl}{d\psi_1}\right)_2 + \beta_2 \left(\frac{dl}{dl_0}\right)_2 \right\}^2 - (11)_2 \left\{ \alpha_2 \left(\frac{dl}{d\psi_1}\right)_2 + \beta_2 \left(\frac{dl}{dl_0}\right)_2 \right\} \left\{ \alpha_2 \left(\frac{dl'}{d\psi_1}\right)_2 + \beta_2 \left(\frac{dl'}{dl_0}\right)_2 \right\} \end{cases}$$

§ 13.

In this equation the constants

$$\alpha_1, \quad \beta_1, \quad \alpha_2, \quad \beta_2,$$

are joined among themselves by alternating relations in such a way that at the limits

$$\psi = \psi_1 \quad \text{and} \quad \psi = \psi$$

δl vanishes, and, when taken inside the limits at

$$\psi = \psi_2,$$

has only one value.

To this corresponds:

$$\alpha_1 \left(\frac{dl}{d\psi_1}\right)_1 + \beta_1 \left(\frac{dl}{dl_0}\right)_1 = 0 \qquad \alpha_2 \left(\frac{dl}{d\psi_1}\right) + \beta_2 \left(\frac{dl}{dl_0}\right) = 0$$

$$\alpha_1 \left(\frac{dl}{d\psi_1}\right)_2 + \beta_1 \left(\frac{dl}{dl_0}\right)_2 = \alpha_2 \left(\frac{dl}{d\psi_1}\right)_2 + \beta_2 \left(\frac{dl}{dl_0}\right)_2,$$

or, putting

$$\alpha_1 = \left(\frac{dl}{dl_0}\right)_1 \left\{ \left(\frac{dl}{d\psi_1}\right)_2 \left(\frac{dl}{dl_0}\right) - \left(\frac{dl}{d\psi_1}\right) \left(\frac{dl}{dl_0}\right)_2 \right\} \omega,$$

$$\beta_1 = - \left(\frac{dl}{d\psi_1}\right)_1 \left\{ \left(\frac{dl}{d\psi_1}\right)_2 \left(\frac{dl}{dl_0}\right) - \left(\frac{dl}{d\psi_1}\right) \left(\frac{dl}{dl_0}\right)_2 \right\} \omega,$$

$$\alpha_2 = - \left(\frac{dl}{dl_0}\right) \left\{ \left(\frac{dl}{d\psi_1}\right)_1 \left(\frac{dl}{dl_0}\right)_2 - \left(\frac{dl}{d\psi_1}\right)_2 \left(\frac{dl}{dl_0}\right)_1 \right\} \omega,$$

$$\beta_2 = \left(\frac{dl}{d\psi_1}\right) \left\{ \left(\frac{dl}{d\psi_1}\right)_1 \left(\frac{dl}{dl_0}\right)_2 - \left(\frac{dl}{d\psi_1}\right)_2 \left(\frac{dl}{dl_0}\right)_1 \right\} \omega.$$

When these values are carried into the equation proposed above, there results:

$$U = (11)_2 \left\{ \alpha_1 \left(\frac{dl}{d\psi_1}\right)_2 + \beta_1 \left(\frac{dl}{dl_0}\right)_2 \right\} \left\{ (\alpha_1 - \alpha_2) \left(\frac{dl''}{d\psi_1}\right)_2 + (\beta_1 - \beta_2) \left(\frac{dl''}{dl_0}\right)_2 \right\}.$$

$$= (11)_2\omega^2 \left\{ \left(\frac{dl}{d\psi_1}\right)_2 \left(\frac{dl}{dl_0}\right) - \left(\frac{dl}{d\psi_1}\right)\left(\frac{dl}{dl_0}\right)_2 \right\} \left\{ \left(\frac{dl}{d\psi_1}\right)_2 \left(\frac{dl}{dl_0}\right)_1 - \left(\frac{dl}{d\psi_1}\right)_1 \left(\frac{dl}{dl_0}\right)_2 \right\}$$
$$\times \left\{ \left(\frac{dl}{d\psi_1}\right)_1 \left(\frac{dl}{dl_0}\right) - \left(\frac{dl}{d\psi_1}\right)\left(\frac{dl}{dl_0}\right)_1 \right\} \left\{ \left(\frac{dl}{d\psi_1}\right)_2 \left(\frac{dl''}{dl_0}\right)_2 - \left(\frac{dl''}{d\psi_1}\right)_2 \left(\frac{dl}{dl_0}\right)_2 \right\}$$

and, putting (chap. IV. § 7):

$$\frac{dl}{d\psi_1} = \frac{\sin l_0 \sin\varphi}{v}; \qquad \frac{dl}{dl_0} = \frac{u}{v}$$

$$U = (11)_2\omega^2 \left(\frac{\sin l_0 \sin\varphi_2}{v_2}\frac{u}{v} - \frac{\sin l_0 \sin\varphi}{v}\frac{u_2}{v_2} \right) \left(\frac{\sin l_0 \sin\varphi_2}{v_2}\frac{u_1}{v_1} - \frac{\sin l_0 \sin\varphi_1}{v_1}\frac{u_2}{v_2} \right)$$
$$\times \left(\frac{\sin l_0 \sin\varphi_1}{v_1}\frac{u}{v} - \frac{\sin l_0 \sin\varphi}{v}\frac{u_1}{v_1} \right) \left\{ \left(\frac{\sin l_0 \sin\varphi_2}{v_2}\right) \left(\frac{u}{v}\right)'_2 - \left(\frac{\sin l_0 \sin\varphi}{v}\right)'_2 \left(\frac{u_2}{v_2}\right) \right\}.$$

If we consider that, when $\psi = \psi_1$ is put, u vanishes, that is,

$$u_1 = 0,$$

then it follows:

$$U = (11)_2\omega^2 \sin^4 l_0 \left\{ \frac{u\sin\varphi_2}{vv_2} - \frac{u_2\sin\varphi}{vv_2} \right\} \left\{ -\frac{u_2\sin\varphi_1}{v_1v_2} \right\} \left\{ \frac{u\sin\varphi_1}{vv_1} \right\}$$
$$\times \left\{ \frac{u_2}{v_2} \left(\frac{\sin\varphi}{v}\right)'_2 - \frac{\sin\varphi_2}{v_2}\left(\frac{u}{v}\right)'_2 \right\}$$
$$= (11)_2\omega^2 \frac{uu_2 \sin^4 l_0 \sin\varphi_1 \sin\varphi_2}{v^2v_1^2v_2^2} \{u\sin\varphi_2 - u_2\sin\varphi\} \frac{\sin\varphi_2^2}{v_2^2} \cdot \left\{ \frac{u}{\sin\varphi} \right\}'_2$$
$$= (11)_2\omega^2 \frac{uu_2 \sin^4 l_0 \sin\varphi_1^2 \sin\varphi_2^2}{v^2v_1^2v_2^4} \{u\sin\varphi_2 - u_2\sin\varphi\} \left\{ -\frac{\cos^2 l}{\cos l_0 \sin^2\varphi_2} \right\},$$
$$U = (11)_2 \sin^4 l_0 \frac{\sin^2\varphi_1 \sin\varphi_2^2 \sin^2\varphi \cos^2 l}{\cos l_0} \frac{1}{v^2}\frac{u_2}{v_1^2}\frac{1}{v_2^4} \left(-\frac{u_2}{\sin\varphi_2} \right) \cdot \frac{u}{\sin\varphi} \left\{ \frac{u}{\sin\varphi} - \frac{u_2}{\sin\varphi_2} \right\}.$$

§ 14.

Up to this point we have considered ψ_2, placed between ψ and ψ_1, but not yet more accurately defined; now ψ_2 seems to have to be confined by narrower limits.

For, since

$$\frac{u}{\sin l_0 \sin\varphi},$$

as we showed above, successively assumes, as the longitude increases, the numbers from 0 to $-\infty$, ψ_2 can be taken so near ψ_1 that ψ is carried to the value ψ_2 before

$$\frac{u}{\sin\varphi}$$

is brought to $-\infty$.

Therefore, since

$$\frac{-u_2}{\sin\varphi_2} > 0$$

is true, U will be either positive or negative according as

$$\frac{u}{\sin\varphi}\left\{\frac{u}{\sin\varphi}-\frac{u_2}{\sin\varphi_2}\right\}$$

is either positive or negative.

Correspondingly, when

$$\frac{u}{\sin\varphi}$$

does not vanish again, U ought to be positive; but when it has advanced beyond 0, U may be either positive or negative.

§ 15.

Hence we state the theorem:

"When some point lying very near a geodetic line is joined by geodetic lines with the points A and B, the sum of the lengths of those lines, if $\psi < \psi'$ is the case, is greater than the length '$A\ B$'; if $\psi > \psi$ is the case, it is either smaller or greater."

CHAPTER V.

§ 1.

Now, however, the shortest of all lines, whose definition we proposed at the beginning of the preceding chapter, must be discussed. For this purpose we shall consider all geodetic lines containing the point $A(l,\omega)$. Their equations are all obtained in turn by allowing in

$$\psi-\omega=\pm\int_{\varphi_1}^{\varphi}d\varphi\cos l_0\frac{\sqrt{1-e^2+e^2\sin^2 l_0\cos^2\varphi}}{1-\sin^2 l_0\cos^2\varphi}$$

(in which the angles φ and φ_1 are known to be defined by the formulae

$$\sin l_0\cos\varphi=\sin l \qquad \sin l_0\cos\varphi_1=\sin l_1$$

) the constant l_0 to take all numbers from 0 to $\frac{\pi}{2}$; when l_0 has been fixed, however, the equation still remains double, since the integral can be taken as either positive or negative, so that two geodetic lines touch the same circle.

§ 2.

There are four genera of these lines. For, when the point Π is carried on any one of them to a neighboring point (l,ψ), one has:

$$\begin{array}{ll} \text{either } l < l_1, & \psi > \omega, \\ \text{or } l > l_1, & \psi > \omega, \\ \text{or } l < l_1, & \psi < \omega, \\ \text{or } l > l_1, & \psi < \omega. \end{array}$$

The equations of the lines of the first and fourth genus take this form:

$$\psi - \omega = \int_{\varphi_1}^{\varphi} \cos l_0 \, d\varphi \, \frac{\sqrt{1 - e^2 + e^2 \sin^2 l_0 \cos^2 \varphi}}{1 - \sin^2 l_0 \cos^2 \varphi}$$

and of the second and third this one:

$$\omega - \psi = \int_{\varphi_1}^{\varphi} \cos l_0 \, d\varphi \, \frac{\sqrt{1 - e^2 + e^2 \sin^2 l_0 \cos^2 \varphi}}{1 - \sin^2 l_0 \cos^2 \varphi}.$$

§ 3.

The nature of all geodetic lines is this: each one contains all the points of the zone of the spheroid enclosed by the circles l_0 and $-l_0$.

ARGUMENT.

It must be shown that any geodetic line whatever cuts every circle of the zone successively at all its points; hence that their longitudes successively assume all numbers lying between 0 and 2Π.

The longitudes of this section are sought from the equation

$$\psi - \omega = \pm \int_{\varphi_1}^{\varphi} \cos l_0 \, d\varphi \, \frac{\sqrt{1 - e^2 + e^2 \sin^2 l_0 \cos^2 \varphi}}{1 - \sin^2 l_0 \cos^2 \varphi}$$

in which φ_1 is constant, while the values of the angle φ are derived from the equation

$$\sin l = \sin l_0 \cos \varphi.$$

Putting, in the first section,

$$\varphi = \varphi_2 \qquad \psi = \psi_2,$$

in the following sections successively the angle becomes

$$\varphi = \varphi_2 + 2\pi \qquad \varphi_2 + 4\pi$$

$$\psi = \omega + \int_{\varphi_1}^{\varphi_2 + 2\pi} \cos l_0 \, d\varphi \, \frac{\sqrt{1 - e^2 + e^2 \sin^2 l_0 \cos^2 \varphi}}{1 - \sin^2 l_0 \cos^2 \varphi}$$

$$\omega + \int_{\varphi_1}^{\varphi_2 + 4\pi} \cos l_0 \, d\varphi \, \frac{\sqrt{1 - e^2 + e^2 \sin^2 l_0 \cos^2 \varphi}}{1 - \sin^2 l_0 \cos^2 \varphi}$$

or $\omega + 4\mathcal{A}(l_0) \qquad \omega + 8\mathcal{A}(l_0)$,

or, referring it to the whole genus:

$$\omega + 4n\mathcal{A}(l_0) - 2h\pi,$$

in which formula n successively assumes the values of all integers, and h represents that integer which is contained in

$$\frac{\psi_2 + 4n\mathcal{A}(l_0)}{2\pi}.$$

Therefore, since

$$\frac{\mathcal{A}(l_0)}{2\pi}$$

is irrational (for the contrary case, which occurs very rarely, must be omitted), ψ successively takes all values lying between 0 and 2π; this was to be demonstrated.

§ 4.

But now, before we undertake to ask which is the shortest of all lines between two points $A(l\psi)$ and $B(l_1\psi_1)$ on the spheroid, it is useful to consider some matters bearing on that question.

In that question let us put l greater than l_1. Since it is now clearly apparent from the preceding theorem that, among the lines issuing from the point A, all contain the point B, it remains to ask more accurately which exists as shorter than the others. In this case it is indeed evident that the line which is shorter than the others can be defined once its genus is known.

ARGUMENT.

For if we consider the course of some geodetic line from the point A, and of many other geodetic lines running near it and inclined less toward the meridian of the point A, we find that the geodetic line is cut by some line all the sooner, the smaller the inclination toward it is. From this it follows that all geodetic lines of the same genus cut it beyond the terminus. But since every geodetic line, as was shown above, is shortest up to the terminus, all the more, when it cuts another geodetic line, must it long since have ceased to be the shortest line.

§ 5.

Before, however, we come nearer to determining the genus, let us propose these two theorems.

I. "The geodetic line l_0 I, if, having started from the point A and not yet having reached the point B, it cuts the meridian of the point A (for example in C), is not the shortest of all lines between A and B."

ARGUMENT.

The geodetic line II, whose negative azimuth is equal to the azimuth of the geodetic line I at A, by symmetry both touches the circle l_0 just as line I does and contains

the point C. Since the length of each of these two lines between A and C is the same, there exist from A to B two equal paths, one of which is geodetic line I itself, the other of which is made up of two parts, namely geodetic line II between A and C and geodetic line I between C and B.

But if the former path between A and B were the shortest of all lines, then certainly the latter also would be the shortest of all lines, which cannot happen, because the latter path is not a geodetic line at the point C.

II. "That geodetic line l_0 of the first or third genus which, drawn from A to B, cuts the circle $-l_0$ twice (one of the places indeed being in C), is not the shortest of all lines between A and B."

The reason for demonstrating this is similar to the preceding one. For with the point Π carried from A on the other geodetic line, which touches the circle l_0, to C, and then on the former from C to B, this whole path is equal to the length AB of geodetic line I.

Thus what we asserted has been demonstrated by the preceding mode of argument.

§ 6.

The line touching the circle l_1 at the point A divides the zone enclosed by the circles l_1 and $-l_1$ into two parts, one of which we choose to name "J", lying inside the geodetic line, the other "E", lying outside the geodetic line.

Therefore, once the force contained in the theorems just proposed has been recognized, it is clear that the shortest of all lines between A and $B(l\psi)$, when B lies in the part "J", is of the first or third genus and has no point of conversion between A and B; when B lies in the part "E", it is of the second or fourth genus and has a point of conversion between A and B; if

$$\psi > \omega,$$

it is of the first or second genus, while if

$$\psi < \omega,$$

it is of the third or fourth genus.

Thus, whatever part of the zone contains the point B, it can be determined in which genus of geodetic lines the shortest of all lines between A and B is to be placed; and indeed the geodetic line which is shortest of all does not cut the circle $-l_0$.

§ 7.

These facts tend to this: if a geodetic line is part of the length of a period, it is shorter than the other geodetic lines between A and B; or, "The geodetic line l_0 is, under this condition, the shortest of all lines between A and B, if the difference of the angles φ belonging to these points does not exceed 180°."

CAPUT VI.

§ 1.

We have seen that the most important moments of the geodetic line can be deduced from the formulae of Chapter II proposed by the illustrious Legendre.

For computing lengths and extensions from l, however, the formulae developed by the illustrious Jacobi in the booklet

„Solution nouvelle d'un problème de Géodésie“

are far more suitable, by passing from elliptic integrals to elliptic functions.

§ 2.

Following him, in place of the angle φ, which is variable by itself, we introduce the variable

$$u = \int_0^{\varphi} \frac{d\varphi}{\sqrt{1 - x^2 \sin^2 \varphi}}$$

and express all quantities by means of its functions. Accordingly, for computing φ we use the formulae arising from the inverse equation of the integral; but for computing the remaining quantities it is necessary to carry these values back to the equations of Chapter II.

Thus there results:

$$\varphi = \operatorname{am} u, \qquad \sin\varphi = Su, \qquad \cos\varphi = Cu, \qquad \sqrt{1 - x^2 \sin^2 \varphi} = \Delta u,$$

$$\sin l = \sin l_0 \, Cu, \qquad \cos l = \cos l_0 \sqrt{1 + \operatorname{tg}^2 l_0 \, S^2 u},$$

$$\operatorname{tg} l = \operatorname{tg} l_0 \frac{Cu}{\sqrt{1 + \operatorname{tg}^2 l_0 \, S^2 u}},$$

$$\sqrt{1 - e^2 \cos^2 l} = \frac{b}{a} \cdot \frac{\Delta u}{x'} = \sqrt{1 - e^2 \cos^2 l_0} \, \Delta(u),$$

$$\operatorname{tg} B = \operatorname{tg} B_0 \frac{Cu}{\sqrt{1 + \operatorname{tg}^2 l_0 \, S^2 u}},$$

$$\sin B = \sin B_0 \frac{Cu}{\Delta u}, \qquad \cos B = \cos B_0 \frac{\sqrt{1 + \operatorname{tg}^2 l_0 \, S^2 u}}{\Delta u},$$

$$\sqrt{1 - e^2 \sin^2 B} = \frac{\cos B}{\cos l} = \frac{x'}{\Delta u}.$$

$$\operatorname{tg} T = \frac{\cot l_0}{Su}, \qquad \sin T = \frac{1}{\sqrt{1+\operatorname{tg}^2 l_0\, S^2u}}, \qquad \cos T = \frac{Su}{\sqrt{\cot^2 l_0 + S^2u}},$$

$$s = \frac{b}{x'}\int_0^u \Delta^2 u\, du,$$

$$\psi = \frac{\cot l_0}{\sin B_0}\int_0^u du\, \frac{\Delta^2 u}{\cot^2 l_0 + S^2 u}.$$

§ 3.

In this way the trigonometric functions of the angles

$$B \qquad l \qquad T$$

are represented by elliptic functions of the quantity u; it remains to determine, by means of its Θ functions, the extension and the longitude.

For this reason the formula for the quantity s is transformed into the following:

$$\frac{x's}{b} = u - \int_0^u x^2 S^2 u\, du$$

in which, after using the formula proposed in the *Fundamenta*, p. 133, and putting

$$\int_0^{\frac{\pi}{2}} \frac{d\varphi}{\sqrt{1-x^2\sin^2\varphi}} = K, \qquad \int_0^{\frac{\pi}{2}} d\varphi\sqrt{1-x^2\sin^2\varphi} = E,$$

one obtains:

$$\frac{2Kx'}{b\pi}\cdot s = \frac{2K}{\pi}u + \frac{2K}{\pi}Zu - u\left(\frac{2K}{\pi} - \frac{2E}{\pi}\right) = \frac{2E}{\pi}u + \frac{2K}{\pi}Zu$$

$$s = \frac{b}{x'}\cdot\frac{E}{K}u + \frac{b}{x'}Zu.$$

And if, in place of the constant E, the constant $\Sigma(l_0)$ is introduced, into which s passes on putting $u = K$; that is, if we put

$$\frac{b}{x'}E. = \Sigma(l_0),$$

then

$$s - \frac{u}{K}\Sigma(l_0) = \frac{b}{x'}Zu.$$

§ 4.

It is more difficult to refer the longitude to Θ functions. For this purpose the method used by the illustrious Jacobi in his booklet

differs a little from that which he proposed in the *Fundamenta*.

Although this discrepancy is so small that, by following the former in our investigation, we introduce nothing artificial.

Accordingly we use a new constant v, defined by the formula

$$v = \int_0^{\frac{\pi}{2}-l} \frac{d\varphi}{\sqrt{1-x'^2\sin^2\varphi}}$$

or by these formulae:

$$S(v,x') = \cos l_0; \qquad C(v,x') = \sin l_0; \qquad \Delta(v,x') = \sin B_0.$$

When these values have been carried into our equations, and after using the formulae proposed in the *Fundamenta*, p. 34, there results:

$$\begin{aligned}
\psi &= \frac{S(v,x')}{C(v,x')\Delta(v,x')}\int_0^u \frac{\Delta^2 u\,du}{S^2u + \frac{S^2(v,x')}{C^2(v,x')}} \\
&= -i\frac{Siv\,Civ}{\Delta iv}\int_0^u \frac{\Delta^2 u\,du}{S^2u - S^2iv} \\
&= i\frac{\Delta(iv-iK')S(iv-iK')}{C(iv-iK')}\int_0^u \frac{\Delta^2 u\,du}{1-x^2S^2u\,S^2(iv-iK')} \\
&= iu\frac{\Delta(iv-iK')S(iv-iK')}{C(iv-iK')} - iuZ(iv-iK') - \frac{i}{2}\log\frac{\Theta(u-iv+iK')}{\Theta(u+iv-iK')} \qquad \text{cf. fund. p. 146.} \\
&= iu\frac{Civ}{Siv\,\Delta iv} - iu\left(\frac{i\pi}{2K} + \frac{Civ\,\Delta iv}{Siv} + Ziv\right) - \frac{i}{2}\log\frac{-i.e.^{\frac{\pi(K'-2iu-2v)}{4K}}H(u-iv)}{i.e.^{\frac{\pi(K'+2iu-2v)}{4K}}H(u+iv)}
\end{aligned}$$

cf. p. 163; cf. f. p. 164.

$$\begin{aligned}
&= iux^2\frac{Siv\,Civ}{\Delta iv} + \frac{\pi u}{2K} - iuZiv - \frac{i}{2}\left\{\pi\frac{K'-2iu-2v}{4K} - \pi\frac{K'+2iu-2v}{4K}\right\} \\
&\quad - \frac{i}{2}\log\frac{H(iv-u)}{H(iv+u)} = -iuZ(iv+K) + \frac{i}{2}\log\frac{H(iv+u)}{H(iv-u)}.
\end{aligned}$$

Putting $u = K$, one has $\psi = \mathcal{A}(l_0)$; therefore

$$\mathcal{A}(l_0) = -iKZ(iv+K) + \frac{\pi}{2},$$

and, referring this to the former form,

$$\psi = \frac{u}{K}\left\{\mathcal{A}(l_0) - \frac{\pi}{2}\right\} + \frac{i}{2}\log\frac{H(iv+u)}{H(iv-u)} = \frac{u\mathcal{A}(l_0)}{K} + \frac{i}{2}\log\frac{\Theta(iv-iK'+u)}{\Theta(iv-iK'-u)}.$$

Although Jacobi showed that this equation can be made to assume a far more elegant form, for our purpose we can nevertheless stand by it.

§ 5.

Since the Θ functions are expanded into very rapidly convergent series, and therefore are computed very easily, it will now be useful also to refer the trigonometric functions of the angles

$$B \qquad l \qquad T$$

to Θ functions.

The most important of the formulae developed in this way are these:

$$\sin l = \sin l_0 \sqrt{\frac{x'}{x}}\, \frac{H(u+K)}{\Theta(u)},$$
$$\sqrt{x} \sin B = \sin B_0 \frac{H(u+K)}{\Theta(u+K)},$$
$$\frac{\cos B}{\cos l} = \sqrt{x'} \frac{\Theta u}{\Theta(u+K)},$$
$$\operatorname{tg} T = \cot l_0 \sqrt{x'}\, \frac{\Theta u}{H(u)}.$$

CAPUT VII.

§ 1.

But in order that the quantities

$$l \qquad s \qquad \psi$$

may be brought to calculation from the equations of the preceding chapter, it is necessary to establish formulae for the constants

$$l_0 \qquad B_0 \qquad \Sigma(l_0) \qquad \mathcal{A}(l_0) \qquad x \qquad x' \qquad K \qquad E \qquad q \qquad v,$$

all of them to be computed from one of them.

§ 2.

Usually the circle that is touched by the curve is defined; only very rarely does one depart from this method. When one of the two periods is known, for example when it is asked: „What circle is touched by the shortest line of all defined by two points of the equator?“ then, for computing l_0, we use only the series of Chapter III.

§ 3.

Likewise, now supposing l_0 known, we can determine the quantities

$$B_0 \qquad \Sigma(l_0) \qquad \mathcal{A}(l_0) \qquad x \qquad x'$$

from the formulae:

$$\begin{aligned}
\sqrt{1-e^2}\ \operatorname{tg} B_0 &= \operatorname{tg} l_0, && \text{Cap. I § 7}\\
\tfrac{2}{\pi}\tfrac{\Sigma(l_0)}{b} &= 1 + o\sin^2 l_0 - \tfrac{3}{4}o^2\sin^4 l_0 + \tfrac{5}{4}o^3\sin^6 l_0 \ldots, && \text{Cap. III § 2}\\
\tfrac{a}{b}\left\{1 - \tfrac{2}{\pi}\mathcal{A}(l_0)\right\} &= 2p\cos l_0 + \left\{1 - 2p + \tfrac{25}{8}p^2\right\}p^2\cos^3 l_0 \ldots, && \text{Cap. III § 4}\\
x &= e\sin B_0, && \text{Cap. II § 9}\\
x' &= \tfrac{\cos B_0}{\cos l_0}.
\end{aligned}$$

§ 4.

It remains to compute the constants

$$q \qquad x \qquad E \qquad v.$$

For this purpose it is useful to apply the following formula:

$$\sqrt{x'} = \frac{1 - 2q + 2q^4 \cdots}{1 + 2q + 2q^4 \cdots} \qquad \text{fund. pag. 184.}$$

Since q is a very small quantity, which indeed, as regards the earth, never exceeds the number „0,0004“, q^4 may be wholly neglected.

Therefore there is:

$$\sqrt{x'} = \frac{1 - 2q}{1 + 2q}$$

or:

$$2q = \frac{1 - \sqrt{x'}}{1 + \sqrt{x'}} = \frac{\sqrt{\cos l_0} - \sqrt{\cos B_0}}{\sqrt{\cos l_0} + \sqrt{\cos B_0}},$$

from which q can easily be computed.

In this matter, since x' is sometimes not accurately defined, we grant that it may happen that q cannot be computed to the extreme digits.

In order that this may not occur, another formula must be explained.

There is:

$$\sqrt{x} = 2\sqrt[4]{q}\,\frac{1 + q^2}{1 + 2q} \qquad \text{fund. pag. 184}$$

$$q = \frac{x^2}{16}\left\{\frac{1 + 2q}{1 + q^2}\right\}^4$$

$$\log q = \log\frac{x^2}{16} - 4\log(1 + q^2) + 4\log(1 + 2q) = \log\frac{x^2}{16} + 8q - 12q^2 + \frac{8}{3}q^3.$$

If q is to be computed more accurately, it is necessary to compute some quantity q_1 approaching q more closely, from the formula

$$\log q_1 = \log\frac{x^2}{16} + \frac{x^2}{2};$$

after this has been done, q is defined from the equation

$$\log q = \log\frac{x^2}{16} + 8q_1 - 12q_1^2 + \frac{8}{3}q_1^3.$$

§ 5.

For computing K we use the formula

$$\sqrt{\frac{2K}{\pi}} = 1 + 2q \qquad \text{fund. p. 184}$$

or

$$K = \frac{\pi}{2}\{1 + 2q\}^2.$$

For computing E, we use the formulae

$$\begin{aligned} \tfrac{2K}{\pi}\tfrac{2K}{\pi} - \tfrac{2K}{\pi}\cdot\tfrac{2E}{\pi} &= 8q + 16q^2 + 32q^3 \ldots, & 135 \\ \tfrac{2K}{\pi}\cdot\tfrac{2K}{\pi} &= 1 + 8q + 24q^2 + 32q^3 \ldots, & 105 \end{aligned}$$

By subtracting one obtains:

$$\frac{2K}{\pi}\cdot\frac{2E}{\pi} = 1 + 8q^2,$$

and by dividing by the preceding equation:

$$\frac{2E}{\pi} = \frac{1 + 8q^2}{(1 + 2q)^2}.$$

This is the other method of computing $\Sigma(l_0)$.

For there is:

$$\Sigma(l_0) = \frac{b}{x'}E \qquad \text{Cap. VI § 3}$$

whence

$$\frac{2}{\pi}\Sigma(l_0) = b\frac{1 + 8q^2}{\{1 - 2q\}^2}.$$

§ 6.

It remains for v to be computed; since this is not explicitly available, it is useful to compute the trigonometric functions related to it of the fraction

$$\frac{iv\pi}{2K}.$$

Accordingly we wish to introduce a new constant A, to be defined by the equation

$$\sin A \cos\frac{iv\pi}{2K} = 1$$

or:

$$\sin\frac{iv\pi}{2K} = i\cot A \qquad \operatorname{tg}\frac{iv\pi}{2K} = i\cos A,$$

from which it is understood that only A has to be computed.

For this purpose

$$\frac{\Delta iv}{\sqrt{x'}} = \frac{1+2q\cos\frac{iv\pi}{K}}{1-2q\cos\frac{iv\pi}{K}} = \frac{1-2q+4q\cos^2\frac{iv\pi}{2K}}{1+2q-4q\cos^2\frac{iv\pi}{2K}}$$
$$= \frac{(1-2q)\sin^2 A+4q}{(1+2q)\sin^2 A-4q}.$$

Or, since

$$\Delta iv = \frac{\Delta(v,x')}{C(v,x')} = \frac{\sin B_0}{\sin l_0} = \frac{a}{b}x' = \frac{1+\varepsilon}{1-\varepsilon}x',$$

there follows

$$(1+\varepsilon)\sqrt{x'}\{(1+2q)\sin^2 A-4q\} = (1-\varepsilon)\{(1-2q)\sin^2 A+4q\}$$
$$(1+\varepsilon)\left\{(1-2q)\sin^2 A-4q\frac{1-2q}{1+2q}\right\} = (1-\varepsilon)\{(1-2q)\sin^2 A+4q\}$$
$$2\varepsilon(1-2q)\sin^2 A = 4q\left\{(1-\varepsilon)+(1+\varepsilon)\frac{1-2q}{1+2q}\right\}$$
$$\sin^2 A = \frac{2q}{\varepsilon}\cdot\frac{1}{1-4q^2}\{1-\varepsilon+2q-2q\varepsilon+1-2q+\varepsilon-2q\varepsilon\}$$
$$\sin^2 A = \frac{4q}{\varepsilon}\cdot\frac{1-2q\varepsilon}{1-4q^2}.$$

§ 7.

Besides this one, it is useful to deduce other formulae for computing the angle A.

There is:

$$\sqrt{x'}\frac{Siv}{Civ} = \frac{\sin\frac{iv\pi}{2K}-q^2\sin\frac{3iv\pi}{2K}}{\cos\frac{iv\pi}{2K}+q^2\cos\frac{3iv\pi}{2K}}$$
$$= i\,\mathrm{tg}\,\frac{iv\pi}{2K}\,\frac{1-q^2\{2\cos\frac{iv\pi}{K}+1\}}{1+q^2\{2\cos\frac{iv\pi}{K}-1\}}$$
$$= i\,\mathrm{tg}\,\frac{iv\pi}{2K}\left\{1-4q^2\cos\frac{iv\pi}{K}\right\} = i\,\mathrm{tg}\,\frac{iv\pi}{2K}\left\{1+4q^2-8q^2\cos^2\frac{iv\pi}{K}\right\}$$
$$= -\cos A\{1-4q^2-2q\varepsilon\}.$$

or, since

$$i\frac{Siv}{Civ} = S(v,x') = \cos l_0 \qquad \text{Cap. VI § 4.}$$
$$\sqrt{x}\cos l_0 = \cos A\{1+4q^2-2q\varepsilon\}$$
$$\cos A = \sqrt{\cos l_0\cos B_0}\{1-4q^2+2q\varepsilon\}.$$

§ 8.

The final method of computing the angle A is obtained from the formula

$$Civ = 2\sqrt[4]{q}\cos\frac{iv\pi}{2K}\sqrt{\frac{x'}{x}}\,\frac{1+q^2\left(2\cos\frac{iv\pi}{K}-1\right)}{1-2q\cos\frac{iv\pi}{K}} \qquad \text{fund. p. 184.}$$

$$= \frac{1}{\sin A} \cdot \frac{1-2q}{1+q^2} \cdot \frac{1-3q^2+4q^2\cos^2\frac{iv\pi}{2K}}{1+2q-4q\cos^2\frac{iv\pi}{2K}}$$
$$= \frac{1}{\sin A} \cdot \frac{1-2q}{1+q^2} \cdot \frac{1-3q^2+q\varepsilon}{1+2q-\varepsilon\frac{1-4q^2}{1-2q\varepsilon}}$$
$$= \frac{\sqrt{x'}}{\sin A} \cdot \frac{1-2q\varepsilon}{1+q^2} \cdot \frac{1+q\varepsilon-3q^2}{1-\varepsilon}.$$

Or, since

$$Civ = \frac{1}{C(v,x')} = \frac{1}{\sin l_0},$$

there results

$$\sin A = \frac{\sqrt{x'}\sin l_0}{1-\varepsilon}\{1-q\varepsilon-4q^2\} = \sqrt{\sin l_0 \sin B_0}\left\{1+\frac{\varepsilon^2}{2}-q\varepsilon-4q^2\right\}.$$

§ 9.

Now, however, it is useful to establish the other formula for computing the longitude of the period. For there is:

$$1-\frac{2}{\pi}\mathcal{A}(l_0) = \frac{2K}{\pi} iZ(iv+K) \qquad \text{cap. VI § 4.}$$
$$= \frac{-4iq\sin\frac{iv\pi}{K}}{1-2q\cos\frac{iv\pi}{K}} = -\frac{8iq\,\mathrm{tg}\,\frac{iv\pi}{2K}\cos^2\frac{iv\pi}{2K}}{1+2q-4q\cos^2\frac{iv\pi}{2K}}$$
$$= \frac{8q\cos A}{1+2q\sin^2 A-2q} = \frac{2\varepsilon\cos A}{\frac{1-2q\varepsilon}{1-2q}-\varepsilon} = 2\varepsilon\frac{1-2q}{1-\varepsilon}\cos A.$$

$$\mathcal{A}(l_0) = \frac{\pi}{2} - \pi\varepsilon\frac{1-2q}{1-\varepsilon}\cos A.$$

CAPUT VIII.

§ 1.

In the sixth chapter we showed that the quantities

$$B, \quad l, \quad T, \quad s, \quad \psi, \quad \varphi,$$

can be determined by functions of one variable u. In order to bring them to calculation, the Θ functions are to be developed into series, proceeding according to the ratio and powers of the constant q and the cosines and sines of multiples of the fraction

$$\frac{\pi u}{2K},$$

which, in the manner of analysts, we denote by „x“.

§ 2.

Using these ratios with the formulae of Chap. VI, we obtain:

$$\sin l = \sin l_0 \frac{1-2q}{1+q^2} \cdot \frac{\cos x + q^2 \cos 3x}{1-2q\cos 2x},$$
$$\sin B = \sin B_0 \frac{1+2q}{1+q^2} \cdot \frac{\cos x + q^2 \cos 3x}{1+2q\cos 2x},$$
$$\frac{\cos B}{\sqrt{x'}\cos l} = \frac{1-2q\cos 2x}{1+2q\cos 2x},$$
$$\operatorname{tg} T = \cot l_0 \frac{1+q^2}{1+2q} \cdot \frac{1-2q\cos 2x}{\sin x - q^2 \sin 3x}.$$

§ 3.

In order to compute the extension, the equation is to be transformed

$$s = \frac{2x}{\pi}\Sigma(l_0) + \frac{b}{x'} Z\frac{2Kx}{\pi} \qquad \text{Chap. VI § 3.}$$

With the formula

$$\frac{2K}{\pi} Z\frac{2Kx}{\pi} = \frac{4q\sin 2x}{1-2q\cos 2x}$$

adjoined, it becomes:

$$s = \frac{2x}{\pi}\Sigma(l_0) + b\frac{4q}{(1-2q)^3} \cdot \frac{\sin 2x}{1-2q\cos 2x} \qquad \text{fund. p. 133.}$$

§ 4.

To compute the longitude, in the formula

$$\psi = -x\frac{2K}{\pi} iZ(iv+K) + \frac{i}{2}\log\frac{H(iv+K)}{H(iv-K)} \qquad \text{Chap. VI § 4.}$$

the H functions are to be expanded in series, and for $Z(iv+K)$ its value from Chap. VI § 9 is to be inserted. Hence one obtains:

$$\psi + 2\varepsilon\frac{1-2q}{1-\varepsilon}x\cos A = \frac{i}{2}\log\frac{\sin\left(\frac{iv\pi}{2K}+x\right) - q^2\sin 3\left(\frac{iv\pi}{2K}+x\right)}{\sin\left(\frac{iv\pi}{2K}-x\right) - q^2\sin 3\left(\frac{iv\pi}{2K}-x\right)}$$
$$= \frac{i}{2}\log\frac{\sin\left(\frac{iv\pi}{2K}+x\right)}{\sin\left(\frac{iv\pi}{2K}-x\right)} + \frac{i}{2}\log\frac{1-q^2\left[2\cos\left(\frac{iv\pi}{K}+2x\right)+1\right]}{1-q^2\left[2\cos\left(\frac{iv\pi}{K}-2x\right)+1\right]}.$$

$$= \frac{i}{2}\log\frac{1+\cot\frac{iv\pi}{2K}\operatorname{tg}x}{1-\cot\frac{iv\pi}{2K}\operatorname{tg}x} + \frac{i}{2}\log\left\{1-2q^2\left[\cos\left(\frac{iv\pi}{K}+2x\right)-\cos\left(\frac{iv\pi}{K}-2x\right)\right]\right\}$$
$$= \frac{i}{2}\log\frac{1-i\frac{\operatorname{tg}x}{\cos A}}{1+i\frac{\operatorname{tg}x}{\cos A}} + 2iq^2\sin\frac{iv\pi}{K}\sin 2x.$$

or, putting

$$\cos A\cdot\operatorname{tg}o = \operatorname{tg}x,$$
$$\psi + 2\varepsilon\frac{1-2q}{1-\varepsilon}x\cos A = o + 4iq^2\operatorname{tg}\frac{iv\pi}{2K}\cos^2\frac{iv\pi}{2K}\sin 2x \qquad \text{C. VII § 6.}$$
$$\psi + 2\varepsilon\frac{1-2q}{1-\varepsilon}x\cos A + \varepsilon q\cos A\sin 2x = \operatorname{arc}\left(\operatorname{tg} = \frac{\operatorname{tg}x}{\cos A}\right).$$

§ 5.

We have now seen that the quantities

$$l, \quad B, \quad T, \quad s, \quad \psi,$$

can always be computed from x, and therefore the whole of our inquiry will turn on the determination of this one quantity. We set down the following three especially important specimens of this fact.

I. Which point of the geodetic line has the latitudes l and B? To compute x from l and B we use the formula

$$\frac{\cos B}{\sqrt{x'}\cos l} = \frac{1-2q\cos 2x}{1+2q\cos 2x}.$$

Solving this equation, with regard to „$2q\cos 2x$“, gives:

$$2q\cos 2x = \frac{\sqrt{x'}\cos l - \cos B}{\sqrt{x'}\cos l + \cos B}$$
$$\cos 2x = \frac{1+\sqrt{x'}}{1-\sqrt{x'}}\cdot\frac{\sqrt{x'}\cos l-\cos B}{\sqrt{x'}\cos l+\cos B}$$
$$= \frac{\sqrt{x'}\{\cos l-\cos B\}+\{x'\cos l-\cos B\}}{\sqrt{x'}\{\cos l-\cos B\}-\{x'\cos l-\cos B\}}.$$
$$\frac{1-\cos 2x}{2} = \frac{\cos B - x'\cos l}{\sqrt{x'}\{\cos l-\cos B\}-\{x'\cos l-\cos B\}}$$
$$\frac{1+\cos 2x}{2} = \sqrt{x'}\frac{\cos l-\cos B}{\sqrt{x'}\{\cos l-\cos B\}-\{x'\cos l-\cos B\}}$$
$$\operatorname{tg}^2 x = \frac{1}{\sqrt{x'}}\frac{\cos B - x'\cos l}{\cos l-\cos B}.$$

II. Which point of the geodetic line l_0 has the extension s?

Dividing the formula

$$s = \frac{2x}{\pi}\Sigma(l_0) + \frac{4qb}{(1-2q)^3} \cdot \frac{\sin 2x}{1 - 2q\cos 2x}$$

by $\frac{2}{\pi}\Sigma(l_0)$ gives

$$x = \frac{\pi s}{2\Sigma} - \frac{4q}{1+8q^2} \cdot \frac{\sin 2x}{1-2q\cos 2x}.$$

From this formula it is understood that x differs only a little from $\frac{\pi s}{2\Sigma}$; and, putting

$$x = \frac{\pi s}{2\Sigma} - 4qv \cdot \sin\frac{\pi s}{\Sigma},$$

v departs only a little from 1. Reducing this value in place of x to our formula gives:

$$v\sin\frac{\pi s}{\Sigma} = (1-8q^2)\frac{\sin\left(\frac{\pi s}{\Sigma} - 8qv\sin\frac{\pi s}{\Sigma}\right)}{1 - 2q\cos\left(\frac{\pi s}{\Sigma} - 8qv\sin\frac{\pi s}{\Sigma}\right)}$$

$$= (1-8q^2)\frac{\sin\frac{\pi s}{\Sigma}\left(1 - 32q^2v^2\sin^2\frac{\pi s}{\Sigma}\right) - \cos\frac{\pi s}{\Sigma}8qv\sin\frac{\pi s}{\Sigma}}{1 - 2q\cos\frac{\pi s}{\Sigma} + 2q\sin\frac{\pi s}{\Sigma}\,8qv\sin\frac{\pi s}{\Sigma}}$$

$$v\left\{1 - 2q\cos\frac{\pi s}{\Sigma} + 16q^2v\cdot\sin^2\frac{\pi s}{\Sigma}\right\} = \left\{1-8q^2\right\}\left\{1 - 32q^2v^2\sin^2\frac{\pi s}{\Sigma} - 8qv\cos\frac{\pi s}{\Sigma}\right\}$$

$$v = 1 - 8q^2 - 16q^2v^2\sin^2\frac{\pi s}{\Sigma} - 6qv\cos\frac{\pi s}{\Sigma}$$

or, putting

$$v = 1 - 6\omega q,$$

$$6\omega = 8q + 6v\cos\frac{\pi s}{\Sigma} + 16qv^2\sin^2\frac{\pi s}{\Sigma} = 8q + 16q\sin^2\frac{\pi s}{\Sigma} + 6\cos\frac{\pi s}{\Sigma} - 36\omega q\cos\frac{\pi s}{\Sigma}$$

and

$$6\omega = 6\cos\frac{\pi s}{\Sigma} + qr,$$

$$r = 8 + 16\sin^2\frac{\pi s}{\Sigma} - 36\cos^2\frac{\pi s}{\Sigma} = 4\left(6 - 13\cos^2\frac{\pi s}{\Sigma}\right).$$

With these values reduced to x, it finally becomes, for the extremum,

$$x = \frac{\pi s}{2\Sigma} - 4q\sin\frac{\pi s}{\Sigma}\left\{1 - 6q\cos\frac{\pi s}{\Sigma} - 4q^2\left(6 - 13\cos^2\frac{\pi s}{\Sigma}\right)\right\}.$$

III. Which point of the geodetic line l_0 has the longitude ψ? To compute x from the equation

$$\psi + 2\varepsilon\frac{1-2q}{1-\varepsilon}x\cos A + \varepsilon q\cos A\sin 2x = \operatorname{arc}\left(\operatorname{tg} = \frac{\operatorname{tg} x}{\cos A}\right)$$

putting

$$\operatorname{tg} y\cos A = \operatorname{tg} t$$

gives:

$$\{\psi + 2\varepsilon t\cos A\} + 2\varepsilon\frac{1-2q}{1-\varepsilon}x\cos A - 2\varepsilon t\cos A + \varepsilon q\cos A\sin 2x = \operatorname{arc}\left(\operatorname{tg} = \frac{\operatorname{tg} x}{\cos A}\right);$$

and, putting $\cos A\operatorname{tg}\{\psi + 2\varepsilon t\cos A\} = \operatorname{tg} y$,

$$2\varepsilon\frac{1-2q}{1-\varepsilon}\cos A(x-y) + 2\varepsilon\frac{1-2q}{1-\varepsilon}y\cos A - 2\varepsilon t\cos A + \varepsilon q\cos A\sin 2x$$

$$= \operatorname{arc}\left(\operatorname{tg} = \frac{\operatorname{tg} x}{\cos A}\right) - \operatorname{arc}\left(\operatorname{tg} = \frac{\operatorname{tg} y}{\cos A}\right) = \operatorname{arc}\left(\operatorname{tg} = \frac{\frac{\operatorname{tg} x}{\cos A} - \frac{\operatorname{tg} y}{\cos A}}{1 + \frac{\operatorname{tg} x}{\cos A}\cdot\frac{\operatorname{tg} y}{\cos A}}\right)$$

$$= \operatorname{arc}\left(\operatorname{tg} = \cos A\frac{\operatorname{tg} x - \operatorname{tg} y}{\cos^2 A + \operatorname{tg} x\operatorname{tg} y}\right) = \operatorname{arc}\left(\operatorname{tg} = \frac{\cos A\sin(x-y)}{\cos^2 A\cos x\cos y + \sin x\sin y}\right).$$

From this equation it is evident that the difference $x - y$, which contains the factor q^2, is a very small quantity; therefore:

$$2\varepsilon\cos A(x-y) + 2\varepsilon\frac{1-2q}{1-\varepsilon}y\cos A - 2\varepsilon t\cos A + \varepsilon q\sin 2x\cos A = \frac{\cos A(x-y)}{\cos^2 A\cos^2 y + \sin^2 y}$$

$$2\varepsilon\frac{1-2q}{1-\varepsilon}y - 2\varepsilon t + \varepsilon q\sin 2x = (x-y)\left\{\frac{1}{1-\sin^2 A\cos^2 y} - 2\varepsilon\right\}$$

$$2\frac{1-2q}{1-\varepsilon}y - 2t + q\sin 2x = (x-y)\left\{\frac{1}{\varepsilon - 4q\cos^2 y} - 2\right\}$$

$$x - y = \left\{2\frac{1-2q}{1-\varepsilon}y - 2t + q\sin 2y\right\}\frac{\varepsilon - 4q\cos^2 y}{1 - 2\varepsilon + 8q\cos^2 y}.$$

CAPUT IX.

§ 1.

After the preceding arguments have set out the relations that hold between the variables of the defined geodetic line, in this last chapter we shall determine the elements themselves from certain given data.

Here let us confine ourselves to solving the question that first presents itself, namely:

„What is the latitude l_0 of the circle touched by the shortest of all lines inside the points A“

„(l_1, φ_1) and B (l_2, ψ_2)?“

This whole question is to be referred to computing l_0 from the formula

$$\psi_2 - \psi_1 = \psi = \int_{\varphi_1}^{\varphi_2} d\varphi\ \cos l_0\frac{\sqrt{1 - e^2 + e^2\sin^2 l_0\cos^2\varphi}}{1 - \sin^2 l_0\cos^2\varphi}$$

putting

$$\sin l_1 = \sin l_0\cos\varphi_1, \qquad \sin l_2 = \sin l_0\cos\varphi_2.$$

Although from the two latter equations $\cos\varphi_1$ and $\cos\varphi_2$ can be determined, nevertheless, because of the periodicity of the trigonometric functions, the angles φ_1 and φ_2 themselves are not determinable.

It is therefore necessary that limits be defined, both those that contain φ_1 and those that contain φ_2. For this reason we assume (which is allowed, since every point of conversion of the geodetic line can be taken as its starting point), first, that between the starting point and point A (taking $l_2 > l_1$) no point of conversion is interposed, and then that between points A and B the starting point is not situated.

It follows that (cf. Chap. II § 8) φ_1 is contained between the limits 0 and π; that φ_2, since $\varphi_2 - \varphi_1 < \pi$ (cf. Chap. V § 7), is contained either between the limits π and 2π, or between 0 and π, according as the point of conversion is situated between the points A and B, or is not; that is, according as

$$\psi > \text{ or } < \int_{l_1}^{l_2} dl\, \frac{\cos l_0}{\cos l} \frac{\sqrt{1 - e^2\cos^2 l}}{\sqrt{\sin^2 l_0 - \sin^2 l}}$$

holds, whose integral, in value, does not depart far from the sum

$$\operatorname{arc}(\cos = \operatorname{tg} l_1 \cot l_0) - \frac{e^2}{2}\cos l_0 \operatorname{arc}\left(\cos = \frac{\sin l_1}{\sin l_0}\right).$$

But, to return to the point from which we turned aside, let us compute l_0. Put

$$\chi = \int_{\varphi_1}^{\varphi_2} d\varphi\, \cos l_0 \frac{1 - \sqrt{1 - e^2 + e^2\sin^2 l_0 \cos^2\varphi}}{1 - \sin^2 l_0 \cos^2\varphi}.$$

Our equation passes into

$$\psi + \chi = \int_{\varphi_1}^{\varphi_2} \frac{\cos l_0\, d\varphi}{1 - \sin^2 l_0 \cos^2\varphi} = \operatorname{arc}\left(\operatorname{tg} = \frac{\operatorname{tg}\varphi_2}{\cos l_0}\right) - \operatorname{arc}\left(\operatorname{tg} = \frac{\operatorname{tg}\varphi_1}{\cos l_0}\right)$$

or, since

$$\operatorname{arc}(\operatorname{tg} = x) = \operatorname{arc}\left(\cos = \frac{1}{\sqrt{1 + x^2}}\right),$$

$$\psi + \chi = \operatorname{arc}\left(\cos = \frac{\cos l_0}{\sqrt{\operatorname{tg}^2\varphi_2 + \cos^2 l_0}}\right) - \operatorname{arc}\left(\cos = \frac{\cos l_0}{\sqrt{\operatorname{tg}^2\varphi_1 + \cos^2 l_0}}\right)$$

or, since

$$\operatorname{arc}(\operatorname{tg} = x) = \operatorname{arc}\left(\cos = \frac{1}{\sqrt{1 + x^2}}\right),$$

$$\psi + \chi = \operatorname{arc}\left(\cos = \frac{\cos l_0}{\sqrt{\operatorname{tg}^2\varphi_2 + \cos^2 l_0}}\right) - \operatorname{arc}\left(\cos = \frac{\cos l_0}{\sqrt{\operatorname{tg}^2\varphi_1 + \cos^2 l_0}}\right)$$

$$= \operatorname{arc}\left(\cos = \frac{\cos l_0}{\sqrt{\frac{1}{\cos^2\varphi_2} - \sin^2 l_0}}\right) - \operatorname{arc}\left(\cos = \frac{\cos l_0}{\sqrt{\frac{1}{\cos^2\varphi_1} - \sin^2 l_0}}\right)$$

$$= \operatorname{arc}\left(\cos = \frac{\cos l_0}{\sqrt{\frac{\sin^2 l_0}{\sin^2 l_2} - \sin^2 l_0}}\right) - \operatorname{arc}\left(\cos = \frac{\cos l_0}{\sqrt{\frac{\sin^2 l_0}{\sin^2 l_1} - \sin^2 l_0}}\right)$$
$$= \operatorname{arc}(\cos = \cot l_0 \operatorname{tg} l_2) - \operatorname{arc}(\cos = \cot l_0 \operatorname{tg} l_1).$$

We can reduce this equation to a more elegant form by using the well-known identity

$$\begin{vmatrix} 1 & \cos x & \cos y \\ \cos x & 1 & \cos(x-y) \\ \cos y & \cos(x-y) & 1 \end{vmatrix}$$

in which, putting

$$x = \cot l_0 \operatorname{tg} l_2, \qquad y = \cot l_0 \operatorname{tg} l_1, \qquad x - y = \psi + \chi,$$

one obtains

$$\begin{vmatrix} 1 & \operatorname{tg} l_2 \cot l_0 & \operatorname{tg} l_1 \cot l_0 \\ \operatorname{tg} l_2 \cot l_0 & 1 & \cos(\psi+\chi) \\ \operatorname{tg} l_1 \cot l_0 & \cos(\psi+\chi) & 1 \end{vmatrix} = \begin{vmatrix} \operatorname{tg}^2 l_0 & \operatorname{tg} l_2 & \operatorname{tg} l_1 \\ \operatorname{tg} l_2 & 1 & \cos(\psi+\chi) \\ \operatorname{tg} l_1 & \cos(\psi+\chi) & 1 \end{vmatrix} = 0,$$

or:

$$\text{I.} \qquad \operatorname{tg}^2 l_0 \sin^2(\psi+\chi) = \operatorname{tg}^2 l_1 + \operatorname{tg}^2 l_2 - 2 \operatorname{tg} l_1 \operatorname{tg} l_2 \cos(\psi+\chi).$$

In this equation l_0 occurs both in „$\operatorname{tg}^2 l_0$“ and in „χ“. But if for the moment we do not wish to define l_0 strictly, but rather a value l_0' which is close to it, then χ may be neglected in the formula, since it contains the very small factor e^2.

Therefore l_0', defined by the formula

$$\text{II.} \qquad \operatorname{tg}^2 l_0' \sin^2 \psi = \operatorname{tg}^2 l_1 + \operatorname{tg}^2 l_2 - 2 \operatorname{tg} l_1 \operatorname{tg} l_2 \cos \psi,$$

differs from l_0 by a quantity that, compared with the eccentricity, is of the second order.

To compute l_0'', which by its value approaches l_0 more closely, χ is not to be wholly thrown away, but account must be taken of the term

$$\frac{e^2}{2} \cos l_0(\varphi_2 - \varphi_1)$$

of the series, progressing according to powers of the square e^2, into which the integral χ can be expanded. Then the value l_0'' defined from this equation,

$$\text{III.} \quad \operatorname{tg}^2 l_0'' \sin^2\left[\psi + \frac{e^2}{2}\cos l_0''(\varphi_2 - \varphi_1)\right] = \operatorname{tg}^2 l_1 + \operatorname{tg}^2 l_2 - 2 \operatorname{tg} l_1 \operatorname{tg} l_2 \cos\left[\psi + \frac{e^2}{2}\cos l_0''(\varphi_2 - \varphi_1)\right],$$

recedes from l_0 by a quantity which, compared with the eccentricity, is of the fourth order.

At first sight one might think serious difficulties obstruct the computation of l_0'' from equation III, since l_0'' is contained not only in $\operatorname{tg}^2 l_0$ but also in

$$\frac{e^2}{2} \cos l_0''(\varphi_2 - \varphi_1)$$

and in the latter place both in $\cos l_0''$ and in the angles φ_2 and φ_1, which are known to be defined by the formulae

$$\sin l_0'' \cos\varphi_2 = \sin l_2, \qquad \sin l_0'' \cos\varphi_1 = \sin l_1.$$

Nevertheless, since

$$\frac{e^2}{2}\cos l_0''(\varphi_2 - \varphi_1)$$

is very small, the quantities contained in it may be computed without error by putting, in place of l_0'', the already known quantity l_0'.

It can happen that l_0' as defined from equation II does not exhibit a value lying near l_0; this occurs precisely when the sum $l_1 + l_2$ is very small and ψ differs not far from π.

Here $\operatorname{tg}^2 l_0$ is represented, as can easily be understood, as the quotient of very small quantities; whence it follows that even the least increment, introduced by adding the integral χ to the quantity ψ, produces a large change in the value of l_0.

If this occurs, for the purpose of computing l_0 one must have recourse to formula II itself; and since, by putting in place of l_0'' the already known value l_0' which is near it, it cannot be reduced to a more elegant form, another procedure must be used, namely putting, for certain quantities, other quantities very close to them.

Since the quantity

$$\psi + \frac{e^2}{2}\cos l_0''(\varphi_2 - \varphi_1)$$

in equation III is not much different from π, we may take

$$\sin\left[\psi + \frac{e^2}{2}\cos l_0''(\varphi_2 - \varphi_1)\right] = \pi - \psi - \frac{e^2}{2}\cos l_0''(\varphi_2 - \varphi_1)$$

$$\cos\left[\psi + \frac{e^2}{2}\cos l_0''(\varphi_2 - \varphi_1)\right] = -1 + \frac{1}{2}\left\{\pi - \psi - \frac{e^2}{2}\cos l_0''(\varphi_2 - \varphi_1)\right\}^2.$$

Thus

$$\operatorname{tg}^2 l_0''\left\{\pi - \psi - \frac{e^2}{2}\cos l_0''(\varphi_2 - \varphi_1)\right\}^2 = \operatorname{tg}^2 l_1 + \operatorname{tg}^2 l_2 + 2\operatorname{tg} l_1 \operatorname{tg} l_2 - \operatorname{tg} l_1 \operatorname{tg} l_2\left\{\pi - \psi - \frac{e^2}{2}\cos l_0''(\varphi_2 - \varphi_1)\right\}^2$$

$$\left\{\operatorname{tg}^2 l_0'' + \operatorname{tg} l_1 \operatorname{tg} l_2\right\}\left\{\pi - \psi - \frac{e^2}{2}\cos l_0''(\varphi_2 - \varphi_1)\right\}^2 = \{\operatorname{tg} l_1 + \operatorname{tg} l_2\}^2.$$

But l_0 does not differ much from $\pi - l_0''$; therefore the sum $\cos\varphi_1 + \cos\varphi_2$ is very small, and hence either $\varphi_2 + \varphi_1$ or $\varphi_2 - \varphi_1$ does not recede far from π. Since this cannot fall in the sum, because the simple angle φ_2 is greater than π (for we have put the longitude ψ as lying near π, and so

$$\psi > \int_{l_1}^{l_2} dl\,\frac{\cos l_0}{\cos l}\,\frac{\sqrt{1 - e^2\cos^2 l}}{\sqrt{\sin^2 l_0 - \sin^2 l}} \qquad \text{cf. p. 35}$$

holds), the difference „$\varphi_2 - \varphi_1$“ lies near π, so that in place of $\varphi_2 - \varphi_1$ one may put π.

Hence:

$$\left\{\operatorname{tg}^2 l_0'' + \operatorname{tg} l_1 \operatorname{tg} l_2\right\}\left\{\pi - \psi - e^2\frac{\pi}{2}\cos l_0''\right\}^2 = \{\operatorname{tg} l_1 + \operatorname{tg} l_2\}^2$$

$$\cos^2 l_1 \cos^2 l_2 \left\{\frac{1}{\cos^2 l_0''} - \frac{\cos(l_1+l_2)}{\cos l_1 \cos l_2}\right\}\left\{\pi - \psi - \frac{e^2}{2}\pi\cos l_0''\right\}^2 = \sin^2(l_1+l_2)$$

$$\cos l_1 \cos l_2 \left\{\frac{\cos l_1 \cos l_2}{\cos^2 l_0''} - \cos(l_1+l_2)\right\}\left\{\pi - \psi - \frac{e^2\pi}{2}\cos l_0''\right\}^2 = \sin^2(l_1+l_2)$$

or, putting

$$\text{in place of } \cos l_2 : \ \cos l_1; \quad \text{in place of } \cos(l_1+l_2) : \ 1; \quad \text{in place of } \sin(l_1+l_2) : \ l_1+l_2,$$

$$\cos^2 l_1 \left\{\frac{\cos^2 l_1}{\cos^2 l_0''} - 1\right\}\left\{\pi - \psi - e^2\frac{\pi}{2}\cos l_0''\right\}^2 = (l_1+l_2)^2$$

$$\cos l_1 \cot T \left\{\pi - \psi - \frac{e^2}{2}\pi\cos l_0''\right\} = l_1 + l_2.$$

In this formula T is known to signify the azimuth of the geodetic line l_0'' at the point A; therefore in l_0'' the formula holds

$$\sin T \cos l_1 = \cos l_0''.$$

Putting this value of $\cos l_0''$ in our equation, there results the formula

$$\cos l_1 \cot T\{\pi - \psi\} - e^2\frac{\pi}{2}\cos^2 l_1 \cos T = l_1 + l_2,$$

from which T and likewise l_0'' can easily be computed.

That equation is sought from these geometrical considerations. The geodetic line l_2, produced beyond the point B until it cuts the circle „$-l_0$“ (for example in C), with the circle $-l_1$ and with the meridian of the point B (which curves we make cut in D) forms a right triangle whose sides are so short that the triangle can be understood and taken as plane:

$$\frac{BD}{CD} = \operatorname{tg} BCD \qquad \text{cf. chap. I § 9.}$$

But, as is easily understood,

$$BD = a\int_{-l_1}^{l_2} dl\sqrt{1 - e^2\cos^2 l}$$

$$CD = a\cos l_1\{2\mathcal{A}(l_0) - \psi\}$$

and the angle BCD makes the complement of T itself. Substituting these into our formula, we obtain

$$\cot T \cos l_1\{2\mathcal{A}(l_0) - \psi\} = \int_{-l_1}^{l_2} dl\sqrt{1 - e^2\cos^2 l}$$

or, neglecting the eccentricity under the sign of integration, and, in the series expansion of the period longitude in Chap. III, taking account only of the first two terms,

$$\cot T \cos l_1\left\{\pi - e^2\frac{\pi}{2}\cos l_0 - \psi\right\} = l_1 + l_2,$$

which was to be demonstrated.

After by some method the value l_0'' has been computed, which differs from l_1 by a very small quantity when regard is had to the eccentricity to the fourth degree, it is easy to compute the latitude l_0 more accurately.

If, expanding the longitude by Taylor's theorem, according to powers of the difference l_0'', we break off the series after the second term, we obtain the formula

$$\psi = \int_{\varphi_1}^{\varphi_2} d\varphi \cos l_0'' \frac{\sqrt{1 - e^2 + e^2 \sin^2 l_0'' \cos^2 \varphi}}{1 - \sin^2 l_0'' \cos^2 \varphi} + (l_0 - l_0'') \frac{\partial \psi}{\partial l_0}$$

or, putting in the values both for the integral and for $\frac{\partial \psi}{\partial l_0}$:

$$\text{V.} \qquad l_0 = l_0'' + \frac{t_2 - t_1 - \psi}{u}$$

in which formula

$$\begin{array}{lrcll}
\text{VI.} & t_2 + 2\varepsilon \frac{1-2q}{1-\varepsilon} x_2 \cos A + \varepsilon q \sin 2x_2 \cos A & = & \operatorname{arc}\left(\operatorname{tg} = \frac{\operatorname{tg} x_2}{\cos A}\right) & \text{cf. chap. VIII § 4} \\
\text{VII.} & t_1 + 2\varepsilon \frac{1-2q}{1-\varepsilon} x_1 \cos A + \varepsilon q \sin 2x_1 \cos A & = & \operatorname{arc}\left(\operatorname{tg} = \frac{\operatorname{tg} x_1}{\cos A}\right) & \\
\text{VIII.} & u \sin B_0 & = & 8q(x_2 - x_1) + (1 - 4q)\{\cot x_2 - \cot x_1\} & \text{cf. chap. VIII § 3.}
\end{array}$$

Now, in order to illustrate the force of the formulae by an example, let

$$B_1 = 48^\circ 43' 5'', \qquad B_2 = 57^\circ 4' 5'', \qquad \psi = 79^\circ 5' 20''.$$

With these premises there results, from formula:

$$a \operatorname{tg} l = b \operatorname{tg} B \quad \text{chap. I § 7}$$

$$l_1 = 48^\circ 37' 22{,}505'' \qquad l_2 = 56^\circ 58' 49{,}647''$$

" "
$$\operatorname{tg}^2 l_0' \sin^2 \psi = \operatorname{tg}^2 l_1 + \operatorname{tg}^2 l_2 - 2 \operatorname{tg} l_1 \operatorname{tg} l_2 \cos \psi \quad \text{chap. IX f. II}$$

$$l_0' = 60^\circ 25' 47''$$

" "
$$\sin l = \sin l_0 \cos \varphi \quad \text{chap. II § 7}$$

$$\varphi_1 = 149^\circ 24' 10'' \qquad \varphi_2 = 195^\circ 25' 33''$$

$$\psi + \frac{e^2}{2} \cos l_0' (\varphi_2 - \varphi_1) = 79^\circ 10' 10''$$

$$\operatorname{tg}^2 l_0'' \sin^2 \left[\psi + \frac{e^2}{2} \cos l_0' (\varphi_2 - \varphi_1)\right] = \operatorname{tg}^2 l_1 + \operatorname{tg}^2 l_2 - 2 \operatorname{tg} l_1 \operatorname{tg} l_2 \cos \left[\psi + \frac{e^2}{2} \cos l_0' (\varphi_2 - \varphi_1)\right]$$

$$l_0'' = 60^\circ 26' 34'' \qquad \text{chap. IX f. III}$$

from formula:

$$a \operatorname{tg} l = b \operatorname{tg} B \qquad \text{chap. I § 7.}$$
$$B_0'' = 60^\circ 31' 31{,}1''$$

" " $\quad x' \cos l_0 = \cos B_0 \qquad \text{chap. II § 9.}$
$$x' = 0{,}9974675491$$

" " $\quad 2q = \frac{1-\sqrt{x'}}{1+\sqrt{x'}} \qquad \text{chap. VII § 4.}$
$$q = 0{,}0003169603$$

" " $\quad \cos A = \sqrt{\cos l_0 \cos B_0}\{1 + 2q\varepsilon - 4q^2\} \qquad \text{chap. VII § 7.}$
$$\log \cos A = 9{,}6925530966$$

" " $\quad \sqrt{x'} \operatorname{tg}^2 x = \frac{\cos B - x' \cos l}{\cos l - \cos B} \qquad \text{chap. VIII, § 5, I.}$

$x_1 = 149^\circ 38' 429{,}65''$; $\qquad x_2 = 195^\circ 25' 10{,}297''$

$$t + 2\varepsilon \frac{1-2q}{1-\varepsilon} x \cos A + \varepsilon q \cos A \sin 2x = \operatorname{arc}\left(\operatorname{tg} = \frac{\operatorname{tg} x}{\cos A}\right) \qquad \text{chap. IX, f. VI and VII.}$$

$t_1 = 130^\circ 10' 28{,}161''$; $\qquad t_2 = 209^\circ 16' 47{,}599''$; $\qquad t_2 - t_1 - \psi = 59{,}438''$

$$u \sin B_0 = 8q(x_2 - x_1) + (1 - 4q)(\cot x_2 - \cot x_1) \qquad \text{chap. IX f. VIII.}$$
$$u = 6{,}125$$
$$l_0 = l_0'' + \frac{t_2 - t_1 - \psi}{u} \qquad \text{chap. IX f. V.}$$
$$l_0 = 60^\circ 26' 43{,}97''.$$

L I F E.

I was born, Paulus Albertus Gordan, attached to the Evangelical faith, at Breslau on 27 April 1837, my father being David, a merchant, my mother Friderica of the Friedenthal family, whom, taken from me by premature death while I was still a boy, I greatly mourn; I greatly rejoice that my father still lives.

Taught the elements of letters by private teachers until my twelfth year, I then attended the Friedrich Gymnasium at Breslau, under the rectorship of Vimmer, who still flourishes there. Two years later, after I had completed the fourth and partly the third class, I was moved by my father to devote myself to trade, and learned, first in Berlin and then in Breslau in my father's affairs, the things necessary for mercantile business; nevertheless I was always drawn to letters by a certain love. At that time I also attended the disciplines of the mercantile school. After I had completed this course, both in order to learn the French language better and to become acquainted with foreign lands, I was sent by my father to Geneva, where I stayed two years.

Having returned from Geneva to my father's business, since from boyhood I had embraced mathematics with the greatest delight, and following this inclination, I was privately instructed in this discipline by Professor Schellbach. Moved by that most excellent man to devote myself wholly to mathematical studies, I left my father's business and resolved to attend the University of Berlin, although I had not yet obtained the certificate of maturity. After I had diligently heard Professor Kummer there for three semesters, I went to Neisse, where, after spending a semester in the first class of the gymnasium, I was furnished with the certificate of maturity.

During the four years completed since that time I devoted myself to mathematical study; first at Breslau, then at Koenigsberg, and then again at Breslau, I attended these lectures:

Mathematics:	Ill. Kummer, Joachimsthal, Richelot, Rosenhain, Schroeter, Galle.
Physics:	Ill. Frankenhein, Neumann.
Astronomy:	Ill. Galle.
Mineralogy:	Ill. Roemer.
Philosophy:	Ill. Braniss.
Politics:	Ill. Tellkampf.
History:	Ill. Junkmann.

To all these men, who have deserved very well of me, I have the greatest thanks and shall always have them.

In the literary contest instituted by the most ample philosophical order of Alma Viadrina,

• On the geodetic line •

I entered the arena, and this little work composed by me on that question was crowned with the prize on the third day before the Nones of August of the preceding year.

THESES. •

I. The division of functions, proposed by the illustrious Cournot into empirical and analytic functions, seems not to be suitable, because empirical functions are not given at all.

II. I contend that the method of infinitesimals is no less accurate than the method of limits.

III. It is of greater interest to inquire what qualities are inherent in a function defined by a differential equation than by what already known functions it can be expressed.

IV. The principles of Democritus down to our own times are the foundations of the doctrine of atoms.